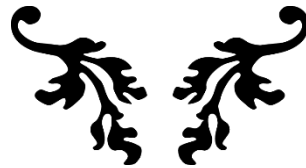




LJ University
University with a Difference



NOTES OF INTERNET OF THINGS



LJ Polytechnic

**Prepared and Compiled by
CE & IT Department**

STUDENT ADVISORY

Dear Students,

Please be informed that the notes provided by the institute offer a concise presentation of the syllabus. While these notes are helpful for an overview and quick revision, We would strongly suggest that you refer to the prescribed textbooks / Reference book for a comprehensive understanding and thorough preparation of all exams and writing in the examination.

Best regards,

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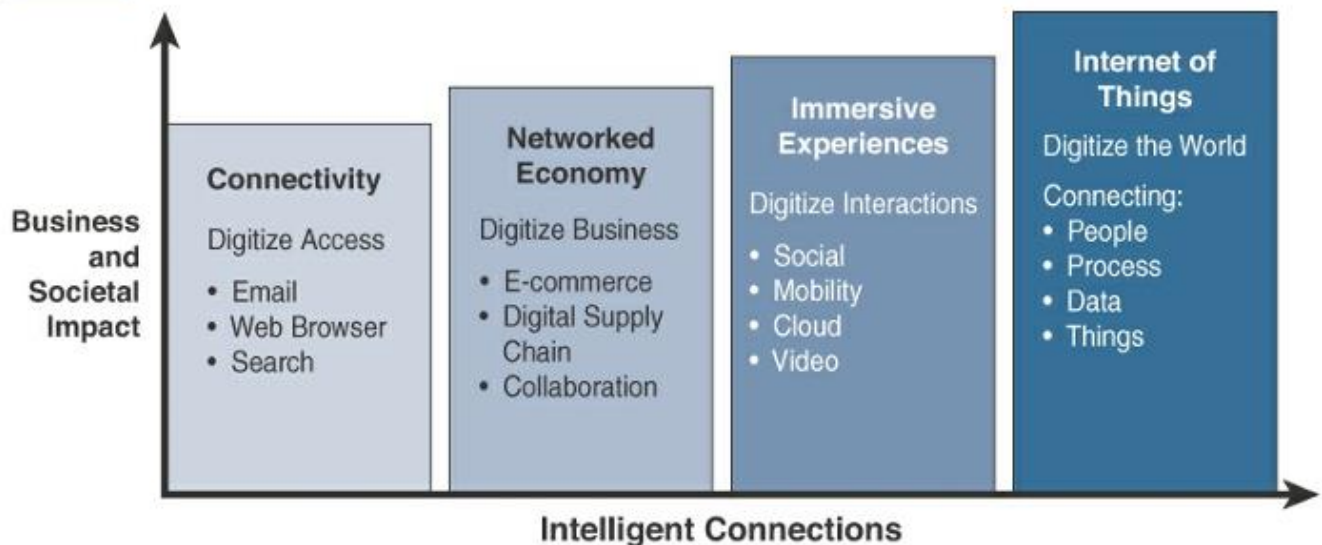
UNIT-1

INTRODUCTION TO INTERNET OF THINGS

1.1 INTRODUCTION OF IOT

Evolutionary Phases of Internet:

As shown in Figure, the evolution of the Internet can be categorized into four phases. Each of these phases have had a profound impact on our society and our lives.



Internet Phase	Definition
Connectivity (Digitize access)	This phase connected people to email, web services, and search so that information is easily accessed.
Networked Economy (Digitize business)	This phase enabled e-commerce and supply chain enhancements along with collaborative engagement to drive increased efficiency in business processes.
Immersive Experiences (Digitize interactions)	This phase extended the Internet experience to encompass widespread video and social media while always being connected through mobility. More and more applications are moved into the cloud.
Internet of Things (Digitize the world)	This phase is adding connectivity to objects and machines in the world around us to enable new services and experiences. It is connecting the unconnected.

IoT stands for the Internet of Things. The term was coined in 1999 by computer scientist Kevin Ashton. There are multiple ways to define IoT, but the basic definition remains the same.

“Internet of Things means a network of physical things (objects) sending, receiving, or communicating information using the Internet or other communication technologies and networks just as computers, tablets, and mobiles do, thus enabling the monitoring, coordinating, or controlling process across the Internet or another data network.”

Another source defines the term IoT as follows:

“Internet of Things refers to the interconnection of physical devices, such as appliances and vehicles, that are embedded with software, sensors, and connectivity which enables these objects to connect and exchange data.”

IoT is not a single technology but it is a combination of technologies and domain knowledge. As a result, engineers from different domains must work together to build a complete IoT product.

The scope and application areas of IoT are very huge. It can be used to build applications for agriculture, the energy sector, defence, embedded applications, education, healthcare, safety and security, home automation, smart cities, manufacturing, transportation, retail industry, media industry, supply chain management, pharmaceuticals, etc.

The term "Things" in the Internet of Things refers to anything and everything in day-to-day life that is accessed or connected through the Internet. A thing in the IoT can be a person with a heart monitor implant, a farm animal with a biochip transponder, or an automobile that has built-in sensors to alert the driver when tire pressure is low. In other words, any other natural or man-made object that can be assigned an Internet Protocol (IP) address and can transfer data over a network can be considered a thing. When something is connected to the internet that means that it can send information, receive information, or both. This ability to send and/or receive information makes things smart, and smarter is better.

For example, in IoT Based Home Automation System following can be considered as “things”:

- Light Control and Automation Devices
- Ventilation devices
- Air Conditioners
- Security Cameras
- Smart Lock
- Smart Switches
- Air Purifiers, etc

In IoT, all the things can be put into 3 categories:

- 1) Sensors that collect information and then send it.
- 2) Computers that receive information and then act on it.
- 3) Things that do both.

- **Sensors that collect information and then send it:**

Sensors can measure temperature, motion, moisture, air quality, light, and almost anything else you can think of. Sensors, when paired with an internet connection, allow us to collect information from the environment which, in turn, helps make better decisions. On a farm, automatically getting information about soil moisture can tell farmers exactly when crops need to be watered. Instead of watering too much or too little (either of which can lead to bad outcomes), the farmer can ensure that crops get exactly the right amount of water. Just as our senses allow us to collect information, sensors allow machines to make sense of their environments.

- **Receiving and Acting on Information:**

We're all very familiar with machines acting on input information. A printer receives a document and then prints it. A garage door receives a wireless signal and the door opens. It's commonplace to remotely command a machine to act. The real power of IoT arises when things can both collect information and act on it.

- **Doing Both:**

Consider the example of farming. The sensors collect information about soil moisture. Now, the farmer could activate the irrigation system or turn it off as appropriate. With IoT-enabled systems, you don't need the farmer for that process. Instead, the irrigation system can automatically act as needed, based on how much moisture is detected. You can take it a step further too. If the irrigation system receives information about the weather from its internet connection, it can also know when it's going to rain and decide not to water the crops when they'll be watered by the rain anyway. And it doesn't stop there. All this information about the soil moisture, how much the irrigation system is watering the crops, and how well the crops grow can be collected and sent to supercomputers in the cloud that run algorithms that analyse all this information, leading to models that could be used to predict future conditions and prevent losses.

Major Components of IoT System:

Major components of IoT devices are:

- 1) **Physical object** with embedded software into hardware.
- 2) **Hardware** consisting of a microcontroller, firmware, sensors, control unit, actuators, and communication module.
- 3) **Communication module:** Software consisting of device APIs and device interface for communication over the network and communication circuit/port(s), and middleware for creating communication stacks using LowPAN, CoAP, LWM2M, IPv4, IPv6, and other protocols.
- 4) **Software** for actions on messages, information, and commands that the devices receive and then output to the actuators, which enable actions such as glowing LEDs, robotic hand movement, etc.

INTERNET OF THINGS

IoT Version	Name / Phase	Key Focus	Core Technologies	Description / Explanation	Real-World Examples
IoT 1.0	Device Connectivity Era	Connecting devices to the internet	RFID, Wi-Fi, Bluetooth, Sensors	First stage where devices were simply connected to the internet to send data. Basic automation and identification systems started.	RFID product tracking, Barcode systems, Early home automation
IoT 2.0	Sensor & M2M Communication era	Machine-to-Machine (M2M) communication	Sensor networks, M2M, Zigbee, Early Cloud	Devices could communicate with each other without human intervention. Remote monitoring and control began.	Smart meters, Wireless sensor networks, Early industrial sensors
IoT 3.0	Cloud & Big Data IoT	Intelligent data processing via Cloud	Cloud computing, Big Data, Mobile Apps, APIs	IoT systems started collecting and analyzing big data. Smart dashboards and mobile apps became common.	Smart wearables, Cloud-based agriculture systems, Connected vehicles
IoT 4.0	Industrial IoT / Automation	Automation + AI + Robotics	AI, ML, Edge Computing, 5G, Cyber-Physical Systems	Used in Industry 4.0: fully automated systems, real-time decision making, predictive maintenance, robotics	Smart factories, Robotic assembly lines, Automated warehouses (Amazon, Tesla)
IoT 5.0	Human-Centric IoT (HC-IoT)	Human + Machine collaboration & emotional intelligence	AI + Human Interaction, Cobots, Emotion-AI, Green IoT	Focus on ethical, sustainable & human-friendly IoT. Systems adapt to human emotions and needs. Humans and robots work together.	Healthcare robots, emotion sensors, Smart eco cities, Assistive devices for elderly
IoT 6.0	Cognitive Brain-IoT Era /	Brain-machine link & quantum intelligence	Quantum Computing, Neural Interfaces, Digital Twins, Metaverse	Future vision where devices understand human thoughts, holographic communication & autonomous decision systems operate globally	Neuralink brain interface, Hologram meetings, Digital twin smart cities, Implanted health IoT

IoT Protocols:

Communication between various things in IoT takes place with the help of various protocols and standards.

Below are some most commonly used protocols in IoT:

- 1) MQTT (Message Queuing Telemetry Transport) Protocol
- 2) CoAP (Constrained Application Protocol)
- 3) DDS (Data Distribution Service) Protocol
- 4) AMQP (Advanced Message Queuing Protocol)
- 5) HTTP/HTTPS
- 6) Bluetooth Protocol
- 7) Wi-Fi Protocol
- 8) Zigbee Protocol
- 9) LoRa WAN (Long Range Wide Area Network) Protocol
- 10) XMPP (eXtensible Messaging and Presence Protocol)

1.2 CHARACTERISTICS OF IoT

1) Connectivity:

Connectivity is an important requirement of the IoT infrastructure. Things of IoT should be connected to the IoT infrastructure. Anyone, anywhere, anytime can connect, this should be guaranteed at all times. IoT can connect through several wireless devices like sensors, mobile phones, trackers, etc.

2) Intelligence and Identity:

The extraction of knowledge from the generated data is very important. For example, a sensor generates data, but that data will only be useful if it is interpreted properly. Each IoT device has a unique identity. This identification helps track the equipment and at times for querying its status.

3) Scalability:

The number of elements connected to the IoT zone is increasing day by day. Hence, an IoT setup should be capable of handling the massive expansion. The data generated as an outcome is enormous and it should be handled appropriately.

4) Dynamic and Self-Adapting:

IoT devices should dynamically adapt to changing contexts and scenarios. For example, consider a camera meant for surveillance. It should be adaptable to work in different conditions and different light situations (morning, afternoon, and night).

5) Architecture:

The IoT architecture is designed to support various devices, technologies, and protocols. Its main task is to confirm whether each connected device does not interfere with the others. This ensures the safety and security of each device's data.

6) Security:

With the increasing number of IoT devices, issues regarding the security of personal data have arisen. There might be a chance of data leakage as a large amount of data is collected, exchanged, and generated. There is a chance of personal data being transferred without approval, which is a matter of concern.

To overcome this challenge, IoT has created networks, systems, and devices wherein privacy is well maintained. Maintaining safety and security is a big dare for IoT. However, it still handles it without any disruption.

7) Self-Configuring:

This is one of the most important characteristics of IoT. IoT devices can upgrade their software under requirements with a minimum of user participation. Additionally, they can set up the network, allowing for the addition of new devices to an already-existing network.

8) Interoperability:

Interoperability is one of the key characteristics of the Internet of Things (IoT). It refers to the ability of different IoT devices and systems to communicate and exchange data with each other, regardless of the underlying technology or manufacturer. To achieve interoperability, IoT devices and systems use standardized communication protocols and data formats.

9) Autonomous Operation:

Autonomous operation refers to the ability of IoT devices and systems to operate independently and make decisions without human intervention.

10) Data Driven:

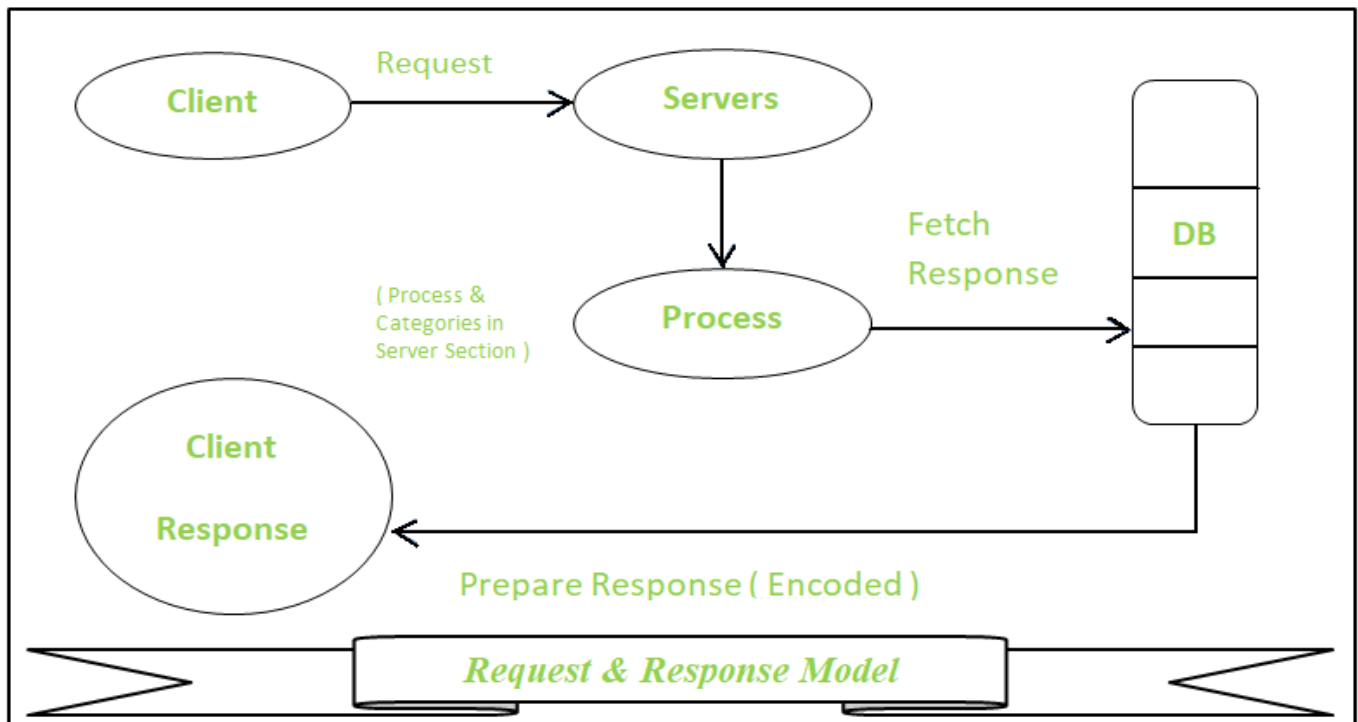
Data-driven is a key characteristic of the IoT. IoT devices and systems collect vast amounts of data from sensors and other sources, which can be analysed, and used to make data-driven decisions.

1.3 MODELS OF IoT

1) Request-Response Model

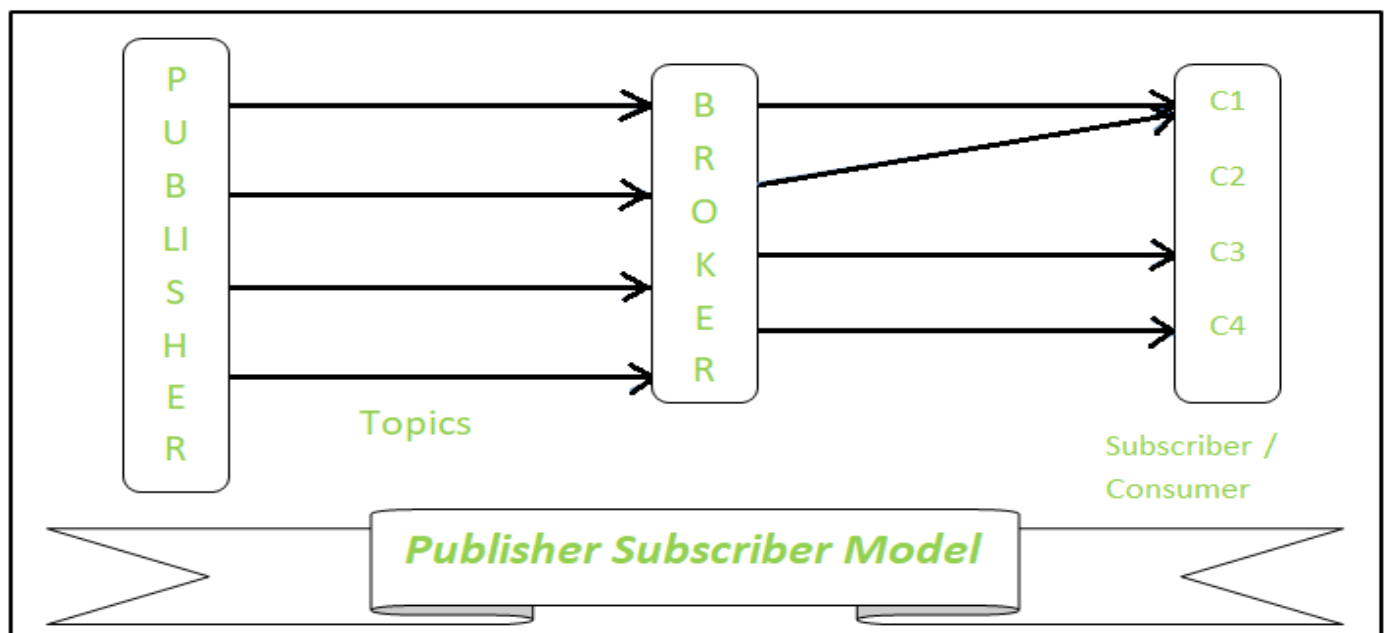
This communication model follows a client-server architecture.

The client, when required, requests the information from the server. This request is usually in the encoded format. This model is stateless since the data between the requests is not retained and each request is independently handled. The server categorizes the request and fetches the data from the database and its resource representation. This data is converted to response and is transferred in an encoded format to the client. The client, in turn, receives the response. On the other hand, in the Request-Response communication model, the client sends a request to the server and the server responds to the request. When the server receives the request, it decides how to respond, fetches the data retrieves resources, prepares the response, and sends it to the client.



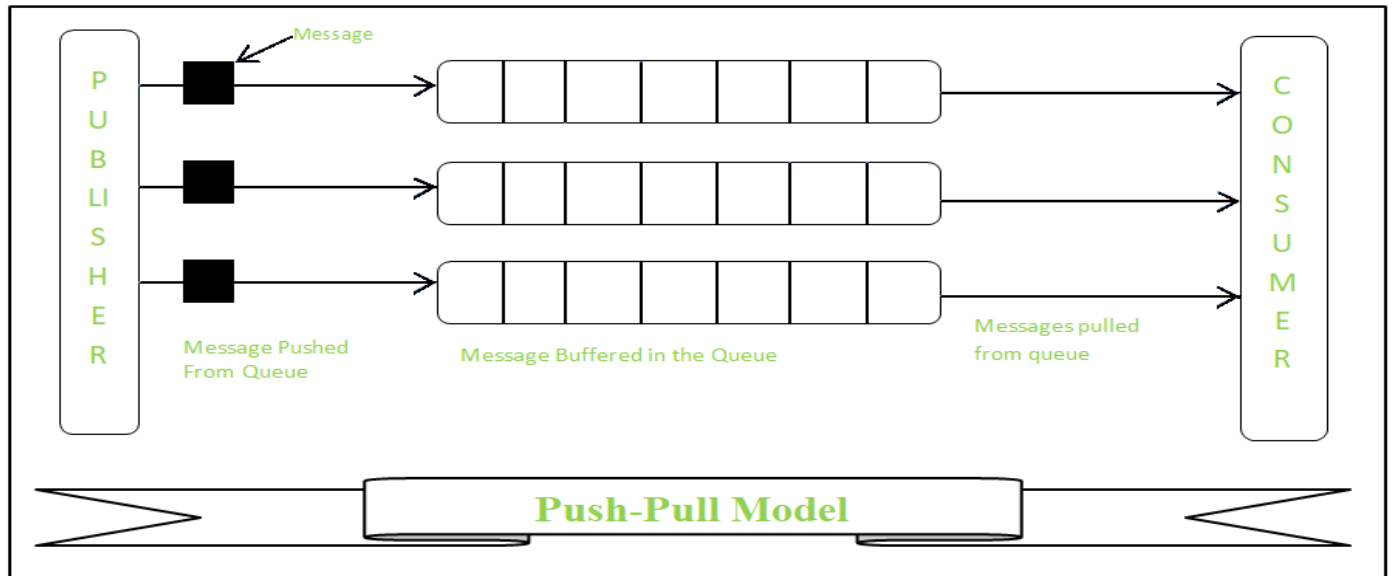
2) Publisher-Subscriber Model

This model comprises three entities: Publishers, Brokers, and Consumers. Publishers are the source of data. It sends the data to the topic which is managed by the broker. They are not aware of consumers. Consumers subscribe to the topics which are managed by the broker. Hence, Brokers responsibility is to accept data from publishers and send it to the appropriate consumers. The broker only has the information regarding the consumer to which a particular topic belongs which the publisher is unaware of.



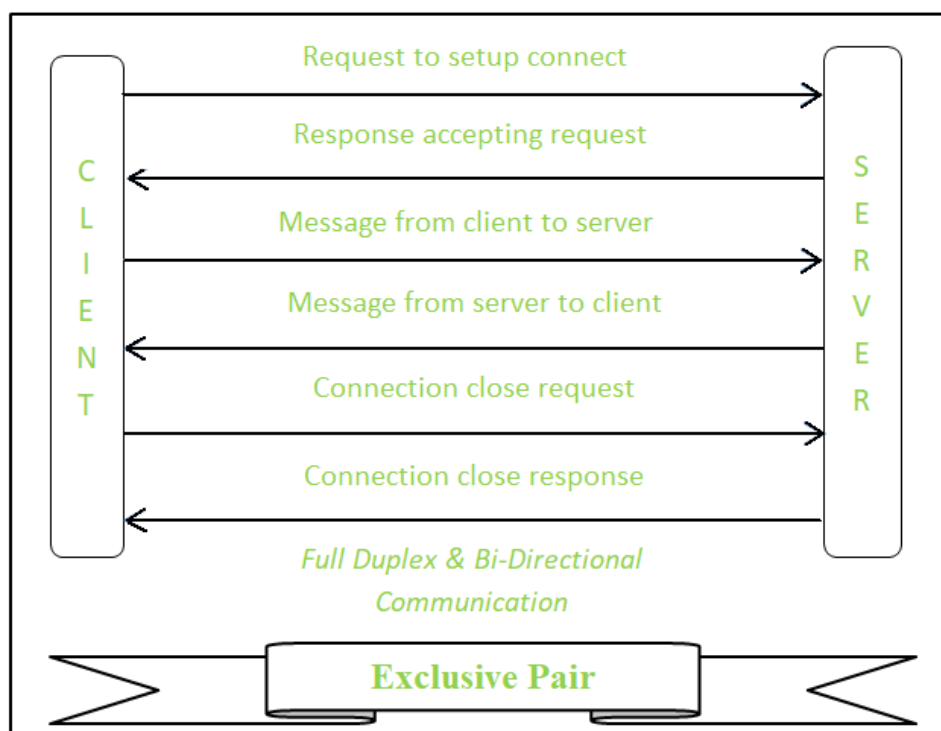
3) Push-Pull Model

The Push-Pull model constitutes data publishers, data consumers and data queues. Publishers and Consumers are not aware of each other. Publishers publish the message/data and push it into the queue. The consumers, present on the other side, pull the data out of the queue. Thus, the queue acts as the buffer for the message when the difference occurs in the rate of push or pull of data on the side of a publisher and consumer. Queues help in decoupling the messaging between the producer and consumer.



4) Exclusive Pair

Exclusive Pair is the bi-directional model, including full-duplex communication among client and server. The connection is constant and remains open till the client sends a request to close the connection. The server has the record of all the connections which has been opened. This is a state-full connection model and the server is aware of all open connections. WebSocket-based communication API is fully based on this model.



1.4 TECHNOLOGY ROADMAP

Creating a technology roadmap specifically for IoT involves considering the unique challenges and opportunities presented by the Internet of Things. Here's a tailored approach for developing an IoT technology Roadmap:

1) Assessment of Current State and Readiness:

- Evaluate existing IoT infrastructure, devices, sensors, and connectivity mechanisms.
- Assess data handling, security measures, and analytics capabilities specific to IoT.

2) Define IoT Objectives and Use Cases:

- Clearly outline the business objectives and use cases for IoT implementation (e.g., predictive maintenance, smart monitoring, supply chain optimization).
- Prioritize use cases based on their potential impact on the business.

3) Stakeholder Engagement and Requirements Gathering:

- Involve stakeholders from various departments to understand their specific IoT needs and challenges.
- Gather requirements to ensure that the IoT roadmap aligns with business goals.

4) Identify Key IoT Technologies:

- Evaluate and select the appropriate IoT technologies (sensors, connectivity protocols, edge computing, platforms) needed to support the defined use cases.
- Consider the scalability, interoperability, security, and data management capabilities of these technologies.

5) Data Management and Analytics Strategy:

- Develop a strategy for managing the vast amount of data generated by IoT devices.
- Plan for real-time or near real-time data processing, analytics, and deriving actionable insights.

6) Security and Privacy Measures:

- Incorporate robust security measures to protect IoT devices, networks, and the data transmitted and stored.
- Implement privacy controls and compliance with relevant regulations.

7) Integration and Interoperability:

- Ensure that IoT systems can integrate with existing IT infrastructure and other applications within the organization.
- Focus on interoperability standards to facilitate seamless communication between devices, and systems.

8) Pilot Projects and Proof of Concepts (POCs):

- Initiate small-scale pilot projects or POCs to validate selected IoT technologies and use cases.
- Gather feedback and refine strategies based on the results.

9) Scaling and Expansion:

- Plan for scaling up successful IoT initiatives across the organization or into new areas.
- Consider the infrastructure, resources, and potential challenges associated with scaling IoT deployments.

10) Continuous Evaluation and Improvement:

- Regularly evaluate the performance, efficiency, and ROI of implemented IoT solutions.
- Incorporate feedback, lessons learned, and technological advancements into the roadmap for continuous improvement.

1.5 BENEFITS OF IoT

1) Increased Efficiency:

- **Automation of Tasks:** IoT enables devices to automate tasks without human intervention, reducing the need for manual input. For example, smart thermostats can automatically adjust heating or cooling based on occupancy and preferences.
- **Real-Time Data Processing:** IoT devices collect and process real-time data, enabling faster decision-making, which increases operational efficiency across industries like manufacturing, healthcare, and logistics.

2) Cost Savings:

- **Energy Efficiency:** Smart devices like lighting systems, thermostats, and appliances can optimize energy use, reducing waste and lowering utility costs.
- **Predictive Maintenance:** IoT sensors can monitor the condition of machinery in real-time, predicting breakdowns before they occur. This reduces the costs associated with unexpected equipment failures and downtime in industries like manufacturing, transportation, and utilities.
- **Reduced Labour Costs:** Automation of routine tasks through IoT reduces the need for human intervention, cutting down on labour costs and increasing productivity.

3) Enhanced Customer Experience:

- **Personalization:** IoT enables companies to collect data on customer behaviour, preferences, and habits. This allows businesses to offer personalized services, such as smart home systems adapting to individual preferences or retail stores offering targeted promotions.
- **Improved Service Delivery:** Connected devices can provide real-time updates on service availability, such as in ride-sharing or delivery services, improving overall customer satisfaction.

4) Better Decision-Making:

- **Data-Driven Insights:** IoT generates vast amounts of data that can be analyzed to gain insights into consumer behaviour, operational performance, and market trends. These insights allow businesses and organizations to make more informed decisions and optimize processes.

- **Real-Time Monitoring:** IoT enables real-time monitoring of critical systems, such as medical devices in healthcare, environmental conditions in agriculture, or energy grids in utilities. This real-time data improves decision-making and responsiveness.

5) Improved Safety and Security:

- **Surveillance and Monitoring:** IoT-connected cameras, sensors, and alarm systems enhance security by enabling remote monitoring and real-time alerts. This is beneficial in areas like home security, workplace safety, and public safety.
- **Health Monitoring:** IoT devices such as wearables or connected medical devices continuously monitor health metrics (e.g., heart rate, blood pressure), providing early warnings about potential health issues and improving patient safety.
- **Disaster Prevention:** IoT sensors can monitor environmental conditions (e.g., temperature, humidity, gas levels), alerting authorities to potential hazards like fires, chemical leaks, or floods.

6) Enhanced Productivity:

- **Supply Chain Optimization:** IoT can track inventory, shipments, and products in real time, leading to optimized supply chain management. This improves stock levels, reduces bottlenecks, and enhances overall productivity in industries like retail and manufacturing.
- **Remote Work & Management:** IoT enables remote management of systems, machines, and devices, allowing businesses to operate more flexibly and employees to work remotely. This is particularly useful for industries like agriculture, energy, and telecommunications.

7) Environmental Benefits:

- **Smart Energy Management:** IoT enables better monitoring and management of energy use in homes, offices, and industrial environments. For example, smart meters and IoT sensors can optimize energy consumption based on real-time demand, reducing waste and emissions.
- **Sustainable Agriculture:** IoT sensors monitor soil moisture, temperature, and crop conditions, allowing farmers to use water and resources more efficiently. This leads to more sustainable farming practices, conserving natural resources.

8) Scalability and Flexibility:

- **Adaptability:** IoT systems are easily scalable, allowing businesses to add new devices and expand their networks without the need for major infrastructure changes.
- **Dynamic Control:** IoT provides the flexibility to manage and control systems remotely from anywhere, adapting to changing business or operational needs in real-time.

9) Innovation and New Business Models:

- **New Revenue Streams:** IoT opens up opportunities for new business models, such as subscription-based services, usage-based pricing, or product-as-a-service models in industries like automotive, healthcare, and manufacturing.

- **Product and Service Innovation:** IoT fosters innovation by enabling new smart products (e.g., wearables, connected home devices) and services that improve user convenience and experience.

10) Integration with Artificial Intelligence (AI):

- **Predictive Analytics:** IoT devices generate vast amounts of data that can be analyzed by AI to predict trends, detect patterns, and optimize operations in real-time. For instance, AI-powered IoT systems can predict equipment failures or consumer demand changes.
- **Automation of Complex Systems:** AI combined with IoT enables fully automated systems where devices not only collect and exchange data but also learn from the data to improve performance. This is particularly useful in autonomous vehicles, smart cities, and industrial applications.

11) Enhanced Mobility and Remote Accessibility:

- **Connected Devices Anywhere:** IoT devices allow for seamless connectivity between mobile applications and remote systems. Users can monitor and control devices (e.g., appliances, machinery, or vehicles) from anywhere using smartphones or tablets.
- **Telemedicine and Remote Health Services:** IoT devices enable remote monitoring and diagnosis for patients, especially in rural or underserved areas, facilitating telemedicine and improving healthcare accessibility.

12) Real-Time Inventory Tracking and Management:

- **Warehouse Management:** IoT devices can monitor inventory levels in real time, ensuring accurate stock tracking and preventing overstocking or understocking situations in warehouses and retail stores.
- **Asset Tracking:** IoT tags or sensors track the location and condition of assets such as machinery, vehicles, or cargo. This is valuable in industries like logistics, retail, and manufacturing.

13) Increased Transparency and Accountability:

- **Supply Chain Visibility:** IoT provides real-time visibility into every aspect of the supply chain, from raw material sourcing to product delivery. This ensures better transparency, accountability, and compliance with regulations.
- **Smart Contracts:** IoT can be integrated with blockchain to create "smart contracts" that automatically execute transactions or trigger actions based on predefined conditions, improving transparency in business transactions.

1.6 IMPACT OF IoT ON THE BUSINESS LANDSCAPE

The Internet of Things (IoT) has had a significant impact on the business landscape across various industries. Following are some key ways in which IoT has influenced and continues to shape businesses:

1) Operational Efficiency:

- **Automation:** IoT enables businesses to automate various processes, reducing the need for manual intervention. This leads to increased operational efficiency and cost savings.

- **Real-time Monitoring:** Businesses can monitor equipment, assets, and processes in real-time, allowing for better decision-making and proactive maintenance.

2) Data-driven Insights:

- **Big Data Analytics:** The vast amount of data generated by IoT devices provides valuable insights. Companies can analyse this data to make informed decisions, optimize processes, and identify new business opportunities.
- **Predictive Analytics:** IoT data allows businesses to predict maintenance needs, demand patterns, and other critical factors, helping them stay ahead of potential issues.

3) Improved Customer Experience:

- **Personalization:** IoT enables businesses to offer personalized products and services based on customer behaviour and preferences.
- **Enhanced Interactivity:** Smart devices and connected systems enhance customer engagement through interactive experiences, creating new ways to connect with consumers.

4) Supply Chain Optimization:

- **Visibility and Traceability:** IoT provides real-time visibility into the supply chain, allowing businesses to track the movement of goods and monitor conditions such as temperature and humidity.
- **Inventory Management:** Smart sensors help optimize inventory levels, reducing waste and ensuring that products are available when needed.

5) Cost Savings and Resource Optimization:

- **Energy Efficiency:** IoT devices can help monitor and control energy consumption, leading to cost savings and a reduced environmental impact.
- **Resource Allocation:** Businesses can optimize the use of resources such as equipment, personnel, and space based on real-time data, minimizing waste and maximizing efficiency.

6) Security and Risk Management:

- **Cybersecurity Measures:** As more devices become interconnected, cybersecurity becomes crucial. Businesses need to implement robust security measures to protect sensitive data and prevent unauthorized access.
- **Risk Mitigation:** IoT enables businesses to identify and address potential risks in real time, reducing the likelihood of disruptions and losses.

7) New Business Models:

- **Subscription Services:** IoT facilitates the shift from traditional product-based business models to service-based models, where companies offer ongoing services and support.
- **Ecosystem Collaboration:** Businesses can create new partnerships and collaborations within the IoT ecosystem, leading to innovative solutions and mutually beneficial relationships.

8) Regulatory Compliance:

- **Data Privacy:** With the increased collection and processing of personal data, businesses must comply with evolving data protection regulations. Implementing privacy measures is essential to maintain trust with customers and meet legal requirements.

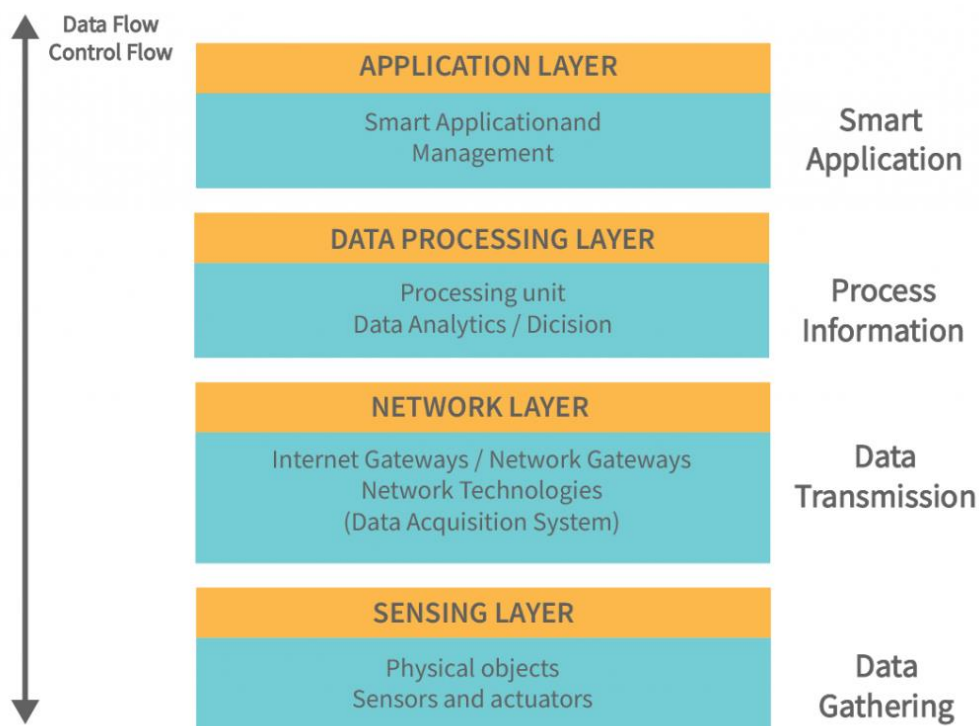
Overall, the integration of IoT technologies has reshaped how businesses operate, offering opportunities for increased efficiency, innovation, and competitiveness in the evolving digital landscape.

UNIT-2

IOT ARCHITECTURE & CHALLENGES

2.1 ARCHITECTURE

Depending upon different application areas of the Internet of Things, it works accordingly as it has been designed/developed. However, it does not have any standard defined architecture of working that is strictly followed universally. The architecture of IoT depends upon its functionality and implementation in different sectors. A four-layer architecture is the standard and most widely accepted format.



1) Sensing Layer

The sensing layer is the first layer of the IoT architecture and is responsible for collecting data from different sources.

This layer includes sensors and actuators that are placed in the environment to gather information about temperature, humidity, light, sound and other physical parameters.

These devices are connected to the network layer through wired or wireless communication protocols.

2) Network Layer

The network layer of an IoT architecture is responsible for providing communication and connectivity between devices in the IoT system.

It includes protocols and technologies that enable devices to connect and communicate with each other and with the wider internet.

Examples of network technologies that are commonly used in IoT include Wi-Fi, Bluetooth, Zigbee and cellular networks such as 4G and 5G.

Additionally, the network layer may include gateways and routers that act as intermediaries between devices and the wider internet and may also include security features such as encryption and authentication to protect against unauthorized access.

3) Data Processing Layer

The data processing layer of IoT architecture refers to the software and hardware components that are responsible for collecting, analysing and interpreting data from IoT devices.

This layer is responsible for receiving raw data from the devices, processing it and making it available for further analysis or action.

The data processing layer includes a variety of technologies and tools such as data management systems, analytics platforms and machine learning algorithms. These tools are used to extract meaningful insights from the data and make decisions based on that data.

Example of a technology used in the data processing layer is a data lake, which is a centralized repository for storing raw data from IoT devices.

4) Application Layer

The application layer of IoT architecture is the topmost layer that interacts directly with the end-user. It is responsible for providing user-friendly interfaces and functionalities that enable users to access and control IoT devices.

This layer includes various software and applications such as mobile apps, web portals, and other user interfaces that are designed to interact with the underlying IoT infrastructure.

It also includes middleware services that allow different IoT devices and systems to communicate and share data seamlessly.

The application layer also includes analytics and processing capabilities that allow data to be analysed and transformed into meaningful insights. This can include machine learning algorithms, data visualization tools, and other advanced analytics capabilities.

2.2 IOT REFERENCE MODEL & ARCHITECTURE

In 2014 the IoTWF architectural committee (led by Cisco, IBM, Rockwell Automation, and others) published a seven-layer IoT architectural reference model. While various IoT reference models exist, the one put forth by the IoT World Forum offers a clean, simplified perspective on IoT and includes edge computing, data storage, and access.

It serves as a common framework to ensure that various elements of IoT ecosystems from sensors and devices to data analytics and user applications can interact seamlessly. The model helps clarify how IoT systems should be designed and how various components should work together.

Using this reference model, we are able to achieve the following:

- Decompose the IoT problem into smaller parts.
- Identify different technologies at each layer and how they relate to one another.
- Define a system in which different parts can be provided by different vendors.
- Have a process of defining interfaces that leads to interoperability.
- Define a tiered security model that is enforced at the transition points between levels.

Levels

- 7 Collaboration & Processes**
(Involving People & Business Processes)
- 6 Application**
(Reporting, Analytics, Control)
- 5 Data Abstraction**
(Aggregation & Access)
- 4 Data Accumulation**
(Storage)
- 3 Edge Computing**
(Data Element Analysis & Transformation)
- 2 Connectivity**
(Communication & Processing Units)
- 1 Physical Devices & Controllers**
(The "Things" in IoT)



Layer 1: Physical Devices and Controllers Layer

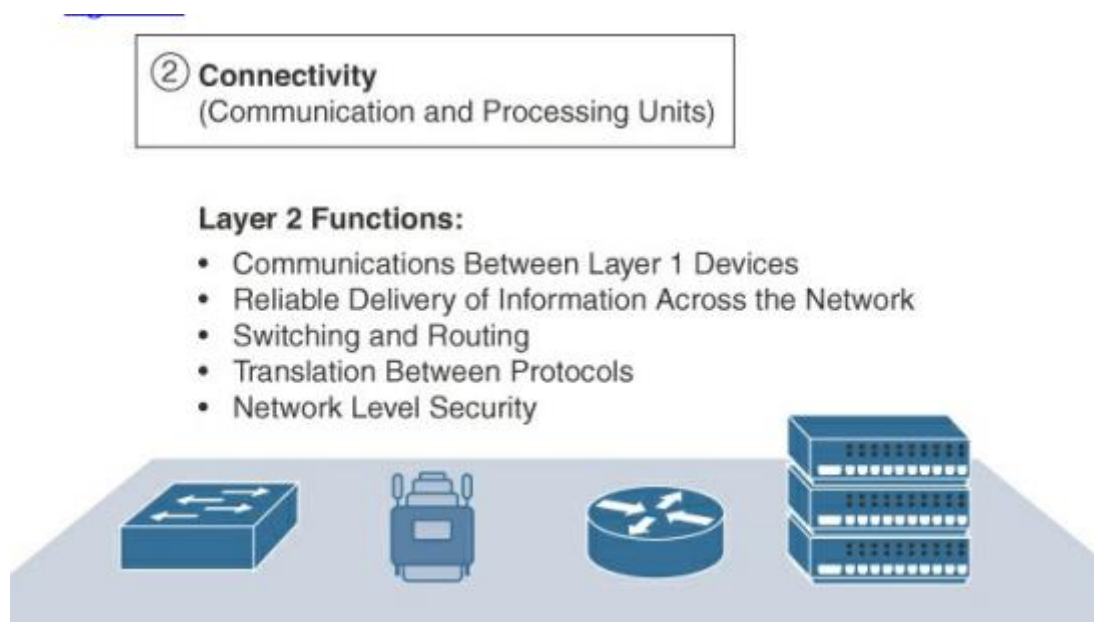
The first layer of the IoT Reference Model is the physical devices and controllers' layer. This layer is home to the "things" in the Internet of Things, including the various endpoint devices and sensors that send and receive information. The size of these "things" can range from almost microscopic sensors to giant machines in a factory. Their primary function is generating data and being capable of being queried and/or controlled over a network.

Examples:

- Sensors (e.g., temperature, humidity, motion sensors).
- Actuators (e.g., motors, valves, smart locks).
- Devices like smart thermostats, cameras, and wearable devices.

Layer 2: Connectivity Layer

In the second layer of the IoT Reference Model, the focus is on connectivity. The most important function of this IoT layer is the reliable and timely transmission of data. More specifically, this includes transmissions between Layer 1 devices and the network and between the network and information processing that occurs at Layer 3 (the edge computing layer).

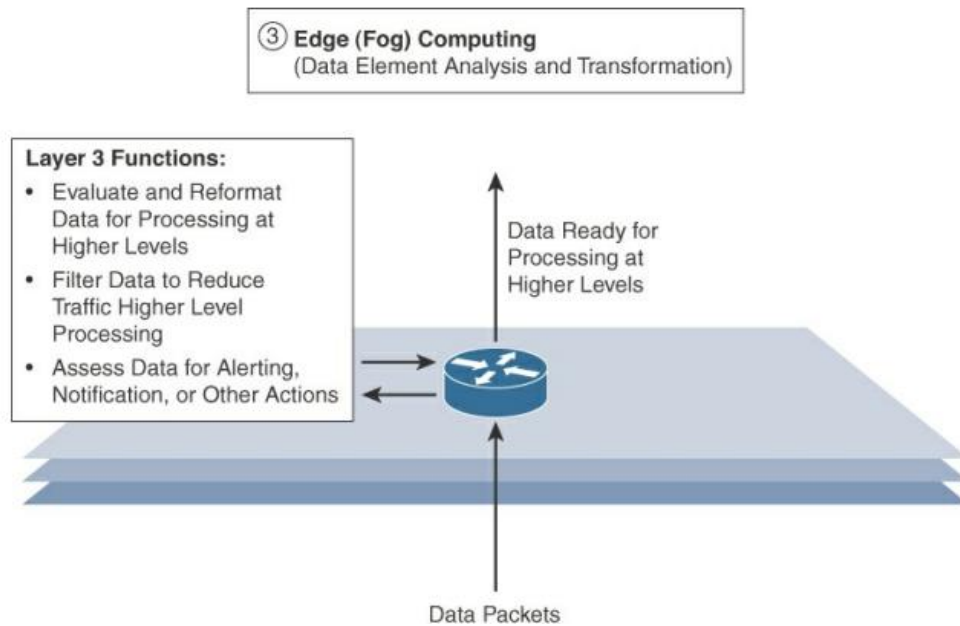


Examples:

- Wireless networks like Wi-Fi, Bluetooth, Zigbee, or cellular 4G/5G networks.
- Wired communication via Ethernet.
- Gateway devices that aggregate data from sensors and forward it to the cloud.

Layer 3: Edge Computing Layer

This layer provides local processing, filtering, and analysis of data closer to the source (edge), rather than sending all raw data directly to the cloud. It helps reduce the latency of decision-making and the bandwidth required to transmit data to the cloud.



Examples:

- Local processing of video streams by smart cameras.
- Edge analytics performed by IoT gateways or routers before sending data to cloud platforms.
- Fog computing platforms that perform local decision-making for smart city applications.

Layer 4: Data Accumulation Layer

This layer is responsible for collecting, normalizing, and aggregating data from the lower layers. It gathers data from various devices and communication systems, ensures it is in a usable format, and prepares it for storage or further analysis in the cloud.

Examples:

- Data brokers and middleware platforms that aggregate data from sensors and edge devices.
- Message queuing services like MQTT brokers or data streams over Kafka.
- Cloud-based data collection services like AWS IoT Core or Microsoft Azure IoT Hub.

Layer 5: Data Abstraction Layer

This layer focuses on organizing, managing, and abstracting the data collected from lower layers to make it usable for analysis and decision-making. It involves processing raw data into formats that can be easily queried and understood, as well as providing data storage and management.

Examples:

- Cloud databases (SQL/NoSQL), distributed file systems like Hadoop or data lakes.
- Storage services like AWS S3 or Google Cloud Storage.
- Middleware that transforms raw IoT data into usable formats for analysis and reporting.

Layer 6: Application Layer

It represents the interface where users and organizations interact with the IoT system. It includes applications and services that deliver value to end-users, whether it's monitoring, control, data visualization, or automation. This layer also focuses on analyzing the data accumulated and processed by the previous layers. It applies data science, artificial intelligence, and machine learning techniques to extract insights, detect patterns, and make decisions based on the data collected from IoT devices.

Examples:

- Predictive analytics in manufacturing, analyzing sensor data to forecast equipment failure.
- Real-time analytics in traffic management systems.
- Machine learning algorithms detecting anomalies in financial transactions or energy usage patterns.
- A mobile app to control smart home devices like lights, thermostats, or security systems.
- Industrial control systems monitoring factory equipment through a dashboard.
- Smart city applications allowing administrators to monitor traffic flows, energy usage, or water systems.

Layer 7: Collaboration and Processes Layer

This layer consumes and shares application information. Collaborating and communication IoT information often requires various steps and this is what makes IoT useful. This layer can change business processes and delivers benefits of IoT.

2.3 FUNCTIONAL, INFORMATION, DEPLOYMENT & OPERATIONAL VIEW

- 1) **Functional View:** Description of what the system does and its main functions.
- 2) **Information View:** Description of the data and information that the system handles.
- 3) **Deployment & Operational View:** Description of the main real-world components of the system such as devices, network routers, servers, etc.

Functional View:

- **Device and Application Functional Group**

The device functional group includes sensing, actuators, processing, storage, and identification components, the sophistication of which depends on the device's capabilities. The application function group contains the standalone application.

- **Communication Functional Group**

It contains the components for end-to-end communication, network communication, and hop-by-hop communication.

- **IoT Service Functional Group**

It corresponds mainly to the service class from the IOT domain model and contains single IOT services exposed by resources hosted on devices or in the network.

- **Virtual Entity Functional Group**

The virtual entity functional group contains functions that support the interactions between users and physical things through virtual entity service.

- **Process Management Functional Group**

It is a collection of functionalities that allows smooth integration of IOT-related Services with the business process.

- **Service Organization Functional Group**

It acts as a communication hub between several other functional groups by composing and orchestrating services of different levels of abstraction.

- **Security Functional Group**

It is responsible for security and privacy matters in IOT-A-compliant IOT systems.

- **Management Functional Group**

It is responsible for the composition and tracking of actions that are involved in the other functional groups.

Information View:

Information view consists of the description of information handled by an IoT system and the way this information is handled in the system i.e. information lifecycle and flow (how information is created, processed, and deleted) and information handling components.

The pieces of information handled by an IoT System:

- Virtual Entity context information, i.e. the attributes (simple or complex) as represented by parts of the IoT Information model.
- IoT Service output itself is another important part of information generated by an IoT system. For example, a Sensor or a Tag Service.
- Virtual Entity descriptions in general, which contain not only the attributes coming from IoT Devices (e.g. ownership information).
- Virtual Entity Associations with other Virtual Entities (e.g. Room #12 is on floor#7)
- **Resource Descriptions:** type of resource (e.g. sensor), identity, associated Services and Devices.
- **Device Descriptions:** device capabilities (e.g. sensors, radios).
- **Descriptions of Composed Services:** the model of how a complex service is composed of simpler services.

- **IoT Business Process Model:** It describes the steps of a business process utilizing other IoT-related services.
- Management information such as state information from operational FCs used for fault/performance purposes, configuration snapshots, reports, membership information, etc.

Deployment and Operational View:

Deployment view and Operational view are very important in addressing how the actual system can be realized by selecting technologies and making them communicate and operate comprehensively.

This view is dependent on the actual use case and requirements.

Let's take an example of the parking lot system

- There are two sensor nodes #1 and #2, each of which is connected to eight car presence sensors.
- They are also connected to the payment stations through wireless or wired communication
- The payment station acts both as a user interface for the device to pay and get a payment receipt as well as a communication gateway that connects the two sensor nodes and payment interface physical devices with the internet through WAN.
- The occupation sign also acts as a communication gateway for the actuator node, and we assume that because of the deployment, a direct connection to the payment station is not feasible.
- The physical gateway devices connect through WAN to the internet and towards a data centre where the parking lot management system software is hosted as one of the virtual machines on a platform as service configurations.
- The two main applications connected to this management system are human-user mobile phone applications and parking operation center applications.

2.4 SIZE & SPACE CONSIDERATIONS IN IOT

Size and Space matter a lot in IoT systems. Today's consumers want everything to be as compact and small as possible. When we consider size and space, there arise two perspectives:

1) IoT Device Perspective

2) IoT Data Perspective

From the IoT device perspective, its size should be as small as possible so that it occupies less space. Further, an IoT device may not be always used as a standalone single device. In most cases, it may be used as a constituent of some other device/system. In such situations also, it is desired to have a small-sized IoT device.

From the IoT data perspective, the space for IoT data storage should be large enough to accommodate the constant streaming of data. The IoT data will be very large over time. So, the space consideration for data storage should be given enough and equal priority in the overall IoT system. For example, IoT wearables are

often small in size; however, they need to be equipped with high performance batteries to perform various types of tasks.

Rural or outdoor areas generally have a large range where the signals need to cover a longer distance to reach to the server or the wireless gateway. This kind of set up requires a GPS or a cellular interface. If the transmission distance is very large, it will require high frequencies as well as high power. If the location is a remote one, then the life of the battery becomes a very important factor.

The operational distance can also be affected by interference with some physical obstacles or with other RF devices. Manufacturers also have many options available in hardware and software relative to the network technology for IoT-based products.

Some IoT devices can be connected directly to the Internet. They may use Ethernet or Wi-Fi. There may be other products that use wireless technologies. But they all will be requiring a 'gateway' to convert their network technology to either Wi-Fi or Ethernet.

Some networks will require antennas and connectors. This requirement will also add to their size. It is a fact that our physical environment is very diverse. This diversity also leads to a similar diverse IoT-based application space, which includes tiny implantable heart rate monitors, oil reservoir diagnosis sensors, HVAC (Heating, Ventilation and Air Conditioning) sensors that have an extremely long lifetime, etc.

2.5 SCALABILITY AND COMPATIBILITY BETWEEN DIFFERENT SMART SENSORS

Introduction to Smart Sensor

A smart sensor is a device that takes input from the physical environment and uses built-in compute resources to perform predefined functions upon detection of specific input and then process data before passing it on.

The smart sensor is also a crucial and integral element in the IoT. IoT technology makes it possible to provide a unique identifier for almost anything and to transmit data from or about those things over the internet or a similar sensor network.

A smart sensor is generally made of a sensor, a microprocessor and wireless communication technology of some kind.

A smart sensor might also include several other components besides the primary sensor. These components can include transducers, amplifiers, excitation control, analog filters, analog-to-digital converters and compensation that provides a built-in correction of less-than-ideal measurements or output. A smart sensor also incorporates software-defined elements that provide functions such as data conversion, digital processing and communication to external devices.

Smart sensors are used mainly for monitoring & control mechanisms in different environments like water level & food monitoring systems, smart grids, traffic monitoring & control, environmental monitoring,

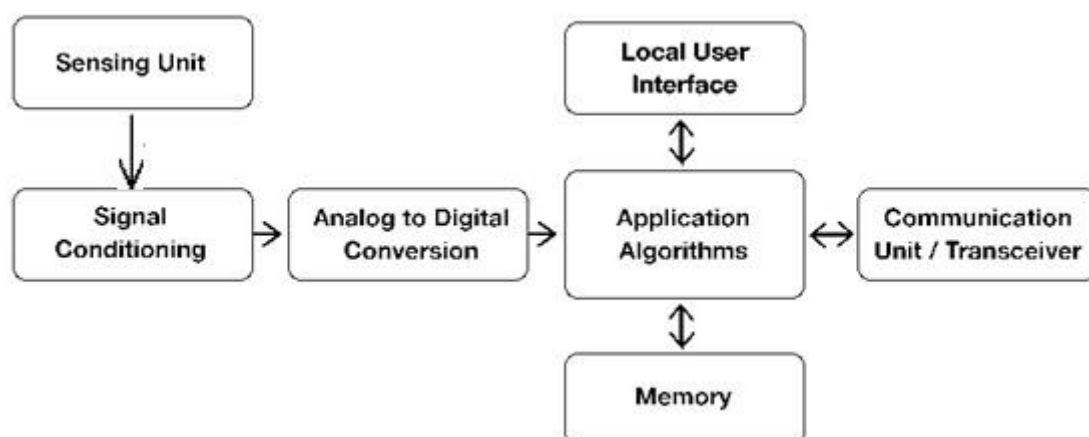
conserving energy in artificial lighting, monitoring of the remote system and fault diagnostics of equipment, transport & logistics, agriculture, telecommunications, industrial applications, animal tracking, etc.

Some of the most commonly used smart sensors are listed as below:

- 1) Temperature Sensors
- 2) Pressure Sensors
- 3) Humidity Sensors
- 4) Infrared Sensors
- 5) Proximity Sensors
- 6) Water Quality Sensors
- 7) Chemical Sensors
- 8) Gas Sensors
- 9) Level Sensors
- 10) Smoke Sensors
- 11) Optical Sensors
- 12) Motion Detection Sensors
- 13) Image Sensors

Smart Sensor Block Diagram

The block diagram of the smart sensor is shown below. This block diagram includes different blocks like sensing unit, signal conditioning, analog to digital conversion, application algorithms, local user interface, memory and communication unit or transceiver.



- **Sensing Unit**

This unit detects the changes in physical parameters & generates electrical signals equivalent to it.

- **Signal Conditioning Unit**

The signal conditioning unit controls the signal to meet the necessities of next-level operations without losing data.

- **Analog to Digital Converter**

ADC converts the signal from analog to digital format & sends it to the microprocessor.

- **Local User Interface**

The local user interface or LUI is a panel-mounted device used to allow building operators to monitor & control system equipment.

- **Application Algorithm**

The signals from smart sensors reach here & process the received data based on the application programs previously loaded here & generate output signals.

- **Memory**

It is used to store media for saving received & processed data.

- **Communication Unit**

The output signals from the application algorithm or microprocessor are transmitted to the main station through the communication unit. This unit also gets command requirements from the key station to execute specific tasks.

Advantages

- 1) These are small in size.
- 2) These sensors are very easy to use, design & maintain.
- 3) The performance level is higher.
- 4) Speed of communication & reliability is higher due to the direct conversion with the processor.
- 5) These sensors can perform self-calibration & self-assessments.
- 6) These sensors can notice issues like switch failures, open coils & sensor contamination.
- 7) These sensors optimize manufacturing processes easily that need changes.
- 8) They can store many systems' data.

Disadvantages

- 1) Smart sensors' reliability is one of the major drawbacks because if they are stolen or get damaged then they can affect a lot of systems badly.
- 2) It needs both sensors & actuators.
- 3) Sensor calibration has to be managed by an external processor.
- 4) High complexity in wired smart sensors, so the cost is also very high.

Difference Smart Sensors and Base Sensors

Parameters	Smart Sensors	Base Sensors
Definition	Sensors with integrated processing, communication, and self-diagnostic capabilities.	Simple sensors that primarily measure and transmit raw data without built-in intelligence.
Data Processing	Can process data locally, perform initial filtering, or aggregation.	No data processing capabilities; only generate and send raw data.
Communication Capabilities	Typically have built-in communication interfaces (e.g., Wi-Fi, Bluetooth, Zigbee) for direct connectivity to networks.	Require external devices (e.g., microcontrollers, gateways) to transmit data.
Integration	Often come with integrated microprocessors or controllers for local decision-making.	Typically lack integrated computational capabilities and need external processing units.
Power Consumption	Higher power consumption due to onboard processing and communication modules.	Lower power consumption since they lack processing and communication modules.
Self-Diagnostics	Can perform self-diagnostics and report sensor health or calibration status.	Do not perform self-diagnostics; external monitoring is required for fault detection.
Cost	More expensive due to added features like processing, communication, and intelligence.	Less expensive because they are simpler devices with basic functionality.
Flexibility	Can be reprogrammed or updated for different applications and functions.	Lack flexibility and are typically designed for a specific measurement or task.
Autonomy	Capable of autonomous decision-making (e.g., activating alarms or actuators) based on data analysis.	Cannot make decisions; only measure and send data for external processing.
Use Cases	Used in advanced applications like smart homes, industrial IoT, healthcare, and autonomous systems.	Used in basic applications where only data collection is required, such as environmental monitoring.
Response Time	May have a slight delay due to internal processing.	Generally faster in data generation as there is no internal processing.

INTERNET OF THINGS

Maintenance	Can report maintenance needs or malfunctions through communication capabilities.	Maintenance or fault detection must be manually monitored by external systems.
Complexity	More complex, requiring configuration and integration with software systems.	Simpler and easier to deploy in basic systems.

UNIT-3

IOT APPLICATIONS IN INDUSTRIES

3.1 BUILDING AND HOME AUTOMATION

Introduction:

IoT-based Home Automation refers to the capacity to use internet-connected equipment to operate household appliances. It might involve pre-programming, complicated heating and lighting systems, as well as alarms and home security controls, all of which are connected via central hub and operated remotely via a smartphone application.

Major Parts of Home Automation System:

A Home Automation System consists of various components that work together to automate and control different aspects of a home environment. Following are the major parts of a typical home automation system:

- 1) **Sensors:** Sensors are devices that detect changes in the environment and convert them into electrical signals. Common types of sensors used in home automation include motion sensors, temperature sensors, humidity sensors, light sensors, door/window sensors, etc. These sensors provide data to the automation system to trigger actions or adjust based on the detected conditions.
- 2) **Actuators:** Actuators are devices that perform physical actions based on commands from the automation system. Examples of actuators used in home automation include light switches, motorized blinds or curtains, smart thermostats, smart locks and smart outlets. Actuators enable the automation system to control various devices and systems within the home.
- 3) **Controller or Hub:** The controller or hub is the central component of the home automation system that manages communication between sensors, actuators and other devices. It collects data from sensors, processes it and sends commands to actuators based on predefined rules or user input. The controller/hub may also provide a user interface for configuring settings, creating automation rules and monitoring the status of connected devices.
- 4) **Network Connectivity:** Home automation devices typically communicate with each other and with the controller/hub over a network. This network may be wired (e.g., Ethernet) or wireless (e.g., Wi-Fi, Zigbee, Z-Wave, Bluetooth). Network connectivity allows remote access to the automation system and enables integration with other smart devices and services.
- 5) **User Interface:** The user interface allows homeowners to interact with the home automation system and control connected devices. This interface may be provided through a mobile app, web portal, touchscreen panel or voice-controlled assistant. Users can use the interface to adjust settings, create automation schedules, receive notifications and monitor the status of their home environment.
- 6) **Automation Rules and Logic:** Automation rules and logic define the behaviour of the home automation system based on specific conditions and triggers. Users can configure rules to automate

routine tasks, such as turning off lights when a room is unoccupied or adjusting thermostat settings based on time of day. The automation system executes these rules automatically, reducing the need for manual intervention.

- 7) Cloud Services:** Some home automation systems may leverage cloud services for data storage, processing and remote access. Cloud services enable homeowners to control their home environment from anywhere with an internet connection and provide additional features such as data analytics, firmware updates and integration with third-party services.

By integrating these major components, a home automation system can enhance convenience, comfort, energy efficiency and security for homeowners, creating a smarter and more connected living environment.

Applications:

- 1) Smart Lighting Control:** Homeowners can remotely control and automate lighting systems using IoT-enabled smart bulbs, switches or dimmers. They can create schedules, adjust brightness levels and even integrate lighting with other smart devices for enhanced security or energy savings.
- 2) Temperature and Climate Control:** IoT thermostats allow users to remotely monitor and control Heating, Ventilation and Air Conditioning (HVAC) systems. They can set temperature schedules, adjust settings based on occupancy or outdoor weather conditions and optimize energy usage for cost savings.
- 3) Security and Surveillance:** IoT-based security systems provide homeowners with remote monitoring and control of cameras, motion sensors, door/window sensors and smart locks. Users can receive real-time alerts, view live video feeds and remotely arm/disarm security systems using their smartphones or tablets.
- 4) Smart Home Appliances:** IoT-enabled appliances such as refrigerators, ovens, washing machines and dishwashers offer advanced features like remote monitoring, control and diagnostics. Users can receive notifications when tasks are completed, adjust settings and even initiate appliance cycles from anywhere.
- 5) Energy Monitoring and Management:** IoT devices equipped with energy monitoring sensors allow homeowners to track and analyse energy consumption patterns in real-time. This data can help identify energy-intensive devices, optimize usage habits and reduce electricity bills.
- 6) Water Leak Detection and Prevention:** IoT-based water leak detection systems use sensors to monitor plumbing fixtures and detect leaks or abnormal water usage. Users receive alerts on their smartphones in case of leaks, allowing for prompt action to prevent water damage and wastage.
- 7) Voice Control and Integration:** Integration with voice-controlled assistants such as Amazon Alexa, Google Assistant or Apple Home Kit allows homeowners to control their IoT devices using voice commands.

- 8) **Automated Window Treatments:** IoT-enabled motorized blinds or curtains can be programmed to open or close automatically based on predefined schedules, sunlight intensity or room occupancy.
- 9) **Health and Wellness Monitoring:** IoT devices equipped with health monitoring sensors can track vital signs, sleep patterns and physical activity levels of household members. This data can be used for personalized health insights, early detection of health issues and remote monitoring of individuals.
- 10) **Integration with Smart Entertainment Systems:** IoT-based home automation systems can integrate with smart entertainment devices such as TVs, speakers and streaming platforms. Users can create custom entertainment scenarios, automate audio/video playback and control multimedia devices seamlessly.

Real Life Examples:

- 1) **Nest Thermostat by Google:** The Nest Thermostat learns users' temperature preferences and automatically adjusts the home's climate to maximize comfort and energy efficiency.
- 2) **Philips Hue Smart Lighting System:** Philips Hue allows users to control lighting from a smartphone app or voice assistants, offering adjustable colors, brightness, and scheduling.
- 3) **Amazon Echo (Alexa):** Amazon Echo is a smart speaker powered by Alexa, Amazon's voice assistant, that enables voice control of various home automation devices.
- 4) **Ring Video Doorbell:** Ring's smart video doorbells allow users to see, hear, and speak to visitors at their door from a smartphone or other connected device.
- 5) **August Smart Lock:** The August Smart Lock allows users to lock and unlock doors using their smartphone and provides remote access control.
- 6) **Samsung SmartThings Hub:** The SmartThings Hub acts as a central hub to connect and control various IoT devices in a home automation setup.
- 7) **iRobot Roomba (Smart Vacuum):** The Roomba is an automated, Wi-Fi-enabled vacuum that uses IoT technology to map home layouts and perform scheduled cleaning.
- 8) **Smart Thermostat:** Similar to Nest, the Ecobee Smart Thermostat optimizes home temperature using room sensors and can be controlled remotely.
- 9) **MyQ Smart Garage Door Opener:** MyQ lets users control their garage door from a smartphone app, allowing remote access and monitoring.
- 10) **Arlo Pro Security Cameras:** Arlo Pro cameras are wire-free security cameras that allow for real-time monitoring and motion detection alerts.
- 11) **Flo by Moen (Smart Water Shutoff):** Flo by Moen monitors water usage, detects leaks, and can automatically shut off the water supply if a leak is detected.
- 12) **Honeywell Home Security System:** Honeywell's security system includes cameras, motion detectors, and alarm systems, all controlled through a central app.

3.2 RETAIL

Introduction:

Today, retail stores are constantly focusing on using the emerging technologies like cloud, mobile, RFID, beacons, etc., to provide connected retail services and better shopping experience to customers. Interestingly, IoT in retail and connected technologies are taking the retail industry by storm. 96% retailers are ready to make changes required to implement the Internet of Things in their stores.

Use of IoT in the smart retail is hugely related to GPS and RFID technologies that help brands in tracking the products throughout entire supply chain process. It gives visibility to the retails for tracking product conditions and movement. It also helps in tracking the location and for predicting an exact delivery time.

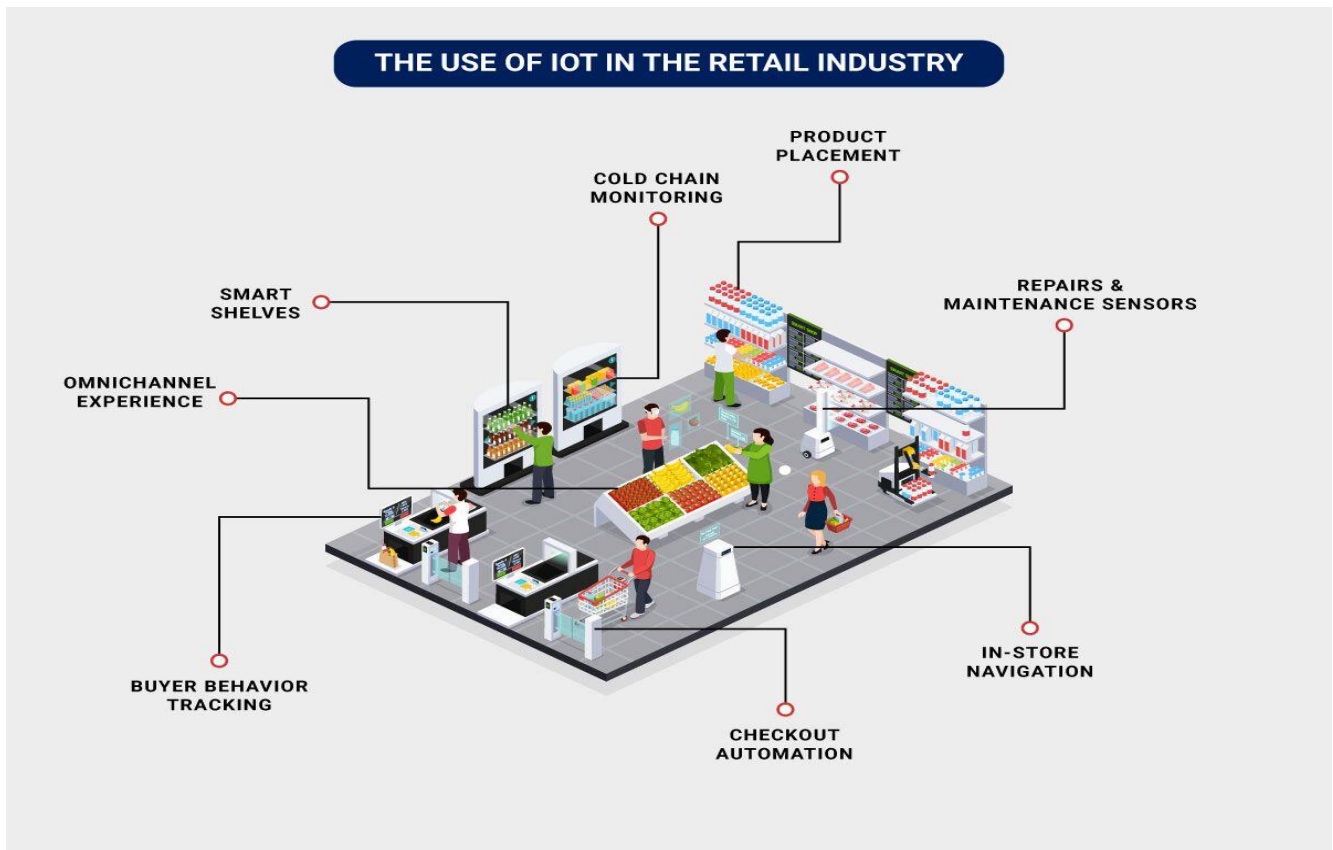
IoT-enabled technologies help in improving various factors such as effective monitoring of store space, proper inventory management and evaluation of customer behaviour. However, data security and high maintenance cost of IoT devices restrain growth of the market globally.

Benefits of IoT in Retail Industry:

- 1) Improves customer experience.
- 2) Improves floor management in-store.
- 3) Allows delivery of coupons and promotions on time.
- 4) Efficient inventory management.
- 5) Optimized supply chain management.

Use of IoT in Retail Industry:

- 1) **Smart Shelves:** IoT-enabled smart shelves automatically track inventory levels and detect when items need to be restocked. They can also display digital price tags and promotions, enabling dynamic pricing strategies and enhancing the in-store experience for customers.
- 2) **Cold Chain Monitoring:** Food spoilage in the grocery industry results in significant losses. Environmental sensors can be installed to track ambient conditions inside a food storage facility, cargo vessels and delivery vehicles to ensure food remains fresh and waste is reduced. They can help obtain precise data, like the temperature at which an item is being stored, how long it is spent in transit, etc. This information can help improve the quality of transportation for the goods in real-time – whether a product is being kept at temperatures too low or too high, avoiding a substantial loss.



- 3) **Product Placement:** Store managers can use the insights collected from sensors and video analytics to identify premium store areas. They can test the placement of different product types in other spots and determine the most profitable layout.
- 4) **Repairs and Maintenance Sensors:** Retailers can install commercial sensors for food storage facilities, parking sites and warehouses to track light, humidity, temperature and air quality. If any vital aspect is detected, an instant alert is triggered. Depending on the software, you can also generate reports to track trends and fluctuations over time and automate safety checks.
- 5) **In-store Navigation:** IoT devices with integrated technologies like Bluetooth, Wi-Fi, magnetic positions and augmented reality, etc., can facilitate in-store navigation to help customers navigate through the store and find the desired product.
- 6) **Checkout Automation:** Stores can use a variety of IoT Technologies, including motion-activated cameras, beacons and shelf-tracking tags, to provide a fully automated, cashier-less in-store shopping experience. Shoppers can walk in, pick the items they want, make the payment online, get a digital receipt on their smartphone and walk out.
- 7) **Buyer Behaviour Tracking:** IoT technology offers retailers valuable insights into buyer behaviour by providing real-time data on customer interactions with physical spaces, products and digital touchpoints IoT helps in tracking buyer behaviour in following way: food traffic analysis, dwell time analysis, customer journey mapping, product interaction tracking, personalised recommendations, queue management, customer feedback collection.

- 8) Omnichannel Experience:** The increasing preference for online shopping has compelled retailers to blend in-store & virtual buying for a seamless experience. IoT devices make this possible by providing consumers with easy access to real-time stock availability in stores. Shoppers can load their carts online and try the products in-store. It ensures a hassle-free in-store experience with minimal time spent.

Examples of IoT in Retail Industry:

The Internet of Things in the retail industry has made significant advancements in retail, transforming traditional shopping experiences and driving operational efficiencies. From cashier-less stores to personalized shopping recommendations, IoT has revolutionized how retailers engage with customers and manage their operations. Here are a few examples of IoT in the retail industry, showcasing its diverse applications and impact:

1) Amazon Go Stores (Seamless Checkout)

- **How it Works:** Amazon Go stores use IoT cameras, sensors, and machine learning to enable a "Just Walk Out" shopping experience. Customers enter the store, pick up items, and leave without waiting in line for a traditional checkout.
- **Impact:** Reduces waiting times, improves the customer experience, and offers valuable data on shopping habits.

2) Zara (Smart Inventory Management)

- **How it Works:** Zara uses RFID tags on clothing items to monitor inventory levels in real-time. The tags provide data to staff about stock levels, making restocking efficient and accurate.
- **Impact:** Streamlines inventory management, reduces stockouts, and enhances in-store operations.

3) Walmart (IoT Robots for Shelf Scanning)

- **How it Works:** Walmart uses shelf-scanning robots in select stores to check for out-of-stock items, pricing errors, and incorrect labels. The robots move around autonomously, scanning shelves and sending data to the central system.
- **Impact:** Improves inventory accuracy, reduces labor costs, and keeps shelves consistently stocked.

4) Walgreens (Smart Coolers)

- **How it Works:** Walgreens installed IoT-enabled smart coolers with digital screens that show advertisements based on demographics like age and gender. The system uses sensors to analyze customers and display relevant ads.
- **Impact:** Offers a targeted advertising experience and boosts impulse purchases, improving marketing effectiveness.

5) Sephora (Smart Mirrors in Fitting Rooms)

- **How it Works:** Sephora uses augmented reality (AR) mirrors that allow customers to try on products virtually. The mirrors let customers see how makeup or other products would look without physically applying them.
- **Impact:** Enhances the in-store experience, reduces product waste, and increases customer engagement.

6) Macy's (Beacons for Personalized Marketing)

- **How it Works:** Macy's uses Bluetooth beacons to send personalized promotions and notifications to customers' smartphones as they move around the store. Beacons recognize customer locations and send targeted messages based on product proximity and interests.
- **Impact:** Boosts customer engagement and drives sales by offering timely, relevant promotions.

7) Kroger (Smart Shelving)

- **How it Works:** Kroger's smart shelves display prices digitally and change dynamically based on factors like stock levels, demand, or promotions. The shelves can also display personalized offers for loyalty program members.
- **Impact:** Reduces manual labor associated with updating shelf labels, enhances in-store experiences, and supports targeted marketing efforts.

8) Decathlon (RFID Self-Checkout)

- **How it Works:** Decathlon, a sporting goods retailer, uses RFID tags on all its products, allowing customers to quickly check out by simply placing items in the checkout area where RFID readers tally them instantly.
- **Impact:** Speeds up the checkout process, improves customer satisfaction, and provides real-time data on stock levels.

9) Lowe's (In-Store Robot Assistant)

- **How it Works:** Lowe's uses an in-store robot named "LoweBot" to assist customers by helping them locate products and providing product information. It uses IoT sensors to navigate the store and interact with customers.
- **Impact:** Enhances customer service, reduces staff workload, and provides a unique in-store experience.

10) Adidas (Connected Stores with IoT Sensors)

- **How it Works:** Adidas stores are equipped with sensors that monitor foot traffic and customer dwell time in specific areas. The data helps optimize store layout, improve staffing, and refine marketing strategies.
- **Impact:** Enhances layout and product placement, increasing the likelihood of purchases and improving overall customer satisfaction.

3.3 MEDIA

Introduction:

As the world becomes ever more integrated, IoT would help companies understand customers and their wants better, in turn allowing them to provide a smoother experience to the consumer. Companies could create detailed profiles for each customer, allowing them to personalise their content better and target ads.

The ability of IoT is to provide users with a personalised experience, advertisers with a much more accurate target and content creators with a bigger sandbox to create a more immersive experience makes it a profitable investment for media companies. During the COVID lockdowns, the soaring stock prices of media houses that adopted the IoT model, like Netflix, Hulu and HBOMax have proved that this is the future. The Internet of Things (IoT) in Media connects the digital and physical worlds.

It has seven levels:

- 1) Device Levels (Data collection and sensors)
- 2) Network Level (Data transmission)
- 3) Data Management and Storage Level
- 4) Analytics and Processing Level
- 5) Application Level (End user experience)
- 6) Automation and Optimization Level
- 7) Feedback and Improvement Level

Examples of IoT in Media Industry:

- 1) **Smart TV Advertising (Personalized Ads):** Smart TVs with IoT capabilities gather data on viewing habits and preferences, allowing media companies to deliver tailored advertisements. These systems use AI to analyze user data and show ads most likely to engage viewers.
- 2) **ESPN (IoT for Real-Time Sports Analytics):** ESPN uses IoT-enabled devices like smart sensors, wearables, and cameras to collect data on athletes' performance in real time. This data is analyzed and broadcasted as real-time stats and highlights during live games.
- 3) **Netflix (Data-Driven Content Recommendations):** Netflix leverages IoT data from various devices to better understand user preferences and optimize recommendations. The IoT devices track user behavior across platforms, allowing Netflix to offer personalized content on any connected device.
- 4) **Augmented Reality (AR) for News Broadcasts:** IoT enables AR applications in live news broadcasting, where sensors and cameras capture real-time data and overlay it on-screen. For example, weather channels use AR to illustrate storm paths and create immersive experiences for viewers.
- 5) **Smart Billboards (Dynamic Content Display):** IoT-powered billboards use sensors to adapt content based on time of day, weather, or viewer demographics. Cameras on billboards can detect approximate age and gender to display ads targeted to the observed audience.

- 6) **Fox Sports (IoT-Powered Fan Experience in Stadiums):** Fox Sports has used IoT technology in stadiums, such as sensors embedded in seating areas and mobile devices, to offer fans live game stats, AR experiences, and personalized replays.
- 7) **Digital Signage in Public Spaces:** IoT-connected digital displays in public spaces adapt content based on weather, time, or crowd density. Sensors help monitor viewer engagement and adjust content to fit audience interests or environmental factors.
- 8) **Wearable Devices for Live Event Broadcasting:** Wearables used by reporters, camera operators, or athletes capture live-streamed audio and video, enabling real-time broadcasting directly from their perspective. This is popular for events such as live concerts, sports, and breaking news coverage.
- 9) **IoT in Film Production (Remote Monitoring of Sets and Equipment):** IoT sensors monitor equipment and environmental conditions on film sets, providing real-time data on lighting, temperature, and humidity. Some systems also monitor the status of critical equipment to prevent failures during shooting.
- 10) **Smart Content Management Systems (CMS):** IoT-powered content management systems use data collected from connected devices to optimize content publishing and scheduling based on audience activity. Some smart CMS can also automate and personalize content across multiple platforms.

Impacts of IoT on Media Industry:

- 1) **Enhanced Content Creation:** IoT devices such as smart cameras and microphones enable media professionals to capture high-quality audio-visual content in various environments. This allows more immersive storytelling and the creation of compelling multimedia experiences.
- 2) **Improved Audience Engagement:** IoT-powered interactive experiences, such as augmented reality (AR) and virtual reality (VR) applications, provide viewers with more engaging and immersive content experiences.
- 3) **Personalized Content Delivery:** IoT devices and platforms collect data on user preferences, viewing habits and demographic information. Leveraging this data, media companies can deliver personalized content recommendations and targeted advertisements, enhancing the overall viewer experience and increasing audience satisfaction.
- 4) **Efficient Content Distribution:** IoT technology enables seamless content delivery across multiple channels and platforms. Smart TVs, streaming devices and set-top boxes equipped with IoT capabilities optimize bandwidth usage and ensure smooth streaming experiences for viewers, regardless of the device or network conditions.
- 5) **Real-Time Analytics and Insights:** IoT sensors collect data on audience behaviour, content consumption patterns and engagement metrics in real-time. Analysing this data provides media companies with valuable insights into audience preferences, content performance and trends that helps in data-driven decision-making and content optimization.

- 6) **Operational Efficiency:** IoT devices and solutions streamline media production workflows and enhance operational efficiency. Asset tracking and management systems powered by IoT technology enables efficient inventory management and maintenance scheduling, reduces downtime and improves resource utilization.
- 7) **Cost Savings:** IoT-enabled solutions optimize resource usage, reduce energy consumption and improve equipment maintenance practices, leading to cost savings for media companies. Predictive maintenance algorithms help to prevent equipment failures and minimize repair costs while energy management systems optimize energy usage and reduce utility expenses.
- 8) **Enhanced Security and Content Protection:** IoT-based surveillance cameras, access control systems and Digital Rights Management (DRM) solutions enhance security and protect valuable media assets. These solutions help prevent unauthorized access, piracy and content theft, ensuring the integrity and security of media content.
- 9) **New Revenue Streams:** IoT technology enables media companies to explore new revenue streams and business models. From subscription-based streaming services to targeted advertising and sponsored content partnerships, IoT-powered media platforms offer opportunities for monetization and revenue growth.

3.4 SUPPLY CHAIN

Introduction:

In the supply chain, Internet of Things devices are an effective way to track and authenticate products and shipments using GPS and other technologies. They can also monitor the storage conditions of products which enhances quality management throughout the supply chain. IoT devices have revolutionized Supply Chain Management (SCM). It's much easier to understand where goods are, how they are being stored and when they can be expected at a specific location.

Some examples of companies successfully using the power of IoT are Amazon, Volvo and Nissan Motor Co. Volvo uses IoT supply chain to track its vehicles' components from several countries and vehicle deliveries to its global customers. Nissan uses IoT supply chain to link its multiple industrial units. Amazon has been using a fleet of IoT-enabled robots to manage warehouse operations by scanning the QR code on packages. The role of IoT in supply chain management is vital for a business to grow.

Benefits of using IoT in Supply Chain Management:

- 1) Reassurance that goods are located where stakeholders say they are, both at rest and in motion.
- 2) Early identification of issues with goods getting lost or delayed.
- 3) Real-time shipment and inventory visibility and tracking.
- 4) Easier supply and demand planning as stakeholders know when they can expect to receive and process goods.

- 5) Better quality management due to keeping raw materials and processed goods in optimal conditions.
- 6) Efficient storage and distribution of products due to the easier location of goods in warehouses.

Examples of IoT Based Supply Chain Management:

- 1) **Walmart:** Walmart utilizes IoT sensors and RFID technology to track inventory levels in its stores in real-time. These sensors automatically monitor stock levels on shelves, triggering alerts when items need to be restocked. This ensures that shelves are consistently well-stocked, reducing out-of-stock situations and improving customer satisfaction.
- 2) **Maersk Line:** Maersk, one of the largest shipping companies globally, employs IoT technology to track and monitor shipping containers. IoT sensors installed in containers provide real-time information on location, temperature, humidity and other environmental conditions. This ensures the integrity of goods during transportation, particularly for perishable items and enables proactive management of any issues that may arise.
- 3) **Amazon:** Amazon utilizes IoT devices and technology in its warehouses to optimize inventory management and order fulfilment processes. IoT-enabled robots and conveyor systems automate tasks such as picking, packing and shipping that reduces the time and labour required for these operations. This enables Amazon to fulfil customer orders quickly and efficiently, even during peak demand periods.
- 4) **UPS:** UPS employs IoT technology to track and monitor its fleet of delivery vehicles. IoT sensors installed in vehicles collect data on factors such as location, speed, fuel consumption, and engine health. This data is analysed in real-time to optimize routes, schedule maintenance and improve fuel efficiency, ultimately reducing costs and enhancing service reliability.
- 5) **Coca-Cola:** Coca-Cola utilizes IoT technology to optimize its vending machine operations. IoT sensors installed in vending machines monitor inventory levels, sales data and machine health metrics in real-time. This enables Coca-Cola to ensure that machines are always stocked with the most popular beverages, anticipate maintenance needs and provide personalized marketing promotions to customers.
- 6) **DHL:** DHL (A global courier, package delivery and express mail service provider company named after its founder Dalsey, Hillblom and Lynn) uses IoT technology to offer end-to-end visibility and tracking of shipments throughout the supply chain. IoT devices installed in packages, vehicles and warehouses enable customers to track their shipments in real-time, from pickup to delivery. This transparency enhances customer satisfaction and enables DHL to provide proactive notifications in case of delays or issues.

Challenges of using IoT Based Supply Chain Management:

- 1) **Skill Gap:** The warehouse staff and vehicle drivers need to be trained and educated about the security practices and the guidelines for using corporate IoT-based supply chain management platforms.
- 2) **Data Storage Challenges:** The large pool of data that IoT systems generate is both a benefit and a challenge. There needs to be enough server power to store and process the collected data. In addition, there must be data governance policies to derive the right conclusions.
- 3) **Security Threats:** Before switching to fully connected systems, you need to have a secure IoT architecture. Vulnerabilities in data can result in outside attacks and leaks. By implementing machine learning (ML) and cryptographic hardware monitoring, managers can decrease security threats.
- 4) **Connectivity Issues:** IoT platforms and devices rely heavily on internet availability and other short distance technologies such as Bluetooth and NFC. As the internet coverage increases and 5G becomes available, this issue will resolve automatically.

3.5 ENVIRONMENTAL MONITORING

Introduction:

IoT environmental monitoring is a process that uses Internet of Things (IoT) technology to collect data about the environment, such as air quality, temperature and humidity levels.

The three main types of environmental monitoring are soil, atmosphere and water. IoT environment monitoring is used in a wide range of industries, from agriculture and forestry to urban planning, energy generation and distribution.

In the **Agricultural Sector**, IoT-based systems are used to monitor crops, soil health, water quality and weather conditions. This information can be used to inform decisions about pest control, fertilisation, irrigation and land management.

IoT-based systems in the **Energy Sector** are used to monitor emissions, air quality and weather conditions. Thus, helping public bodies, environmental agencies and companies to monitor and act to reduce negative environmental impact.

In **Urban Planning**, IoT-based systems can be used to monitor traffic congestion or air pollution levels in smart cities. This data can be used to inform decisions about how to reduce the environmental impacts of future urban development.

IoT based Environmental Monitoring is used for air quality monitoring, water quality monitoring, energy monitoring, commercial farming, toxic gas detection, animal conservation.

These IoT-based systems can be used to detect issues in the environment that are largely invisible, normalised or taken for granted.

Benefits of using IoT based Environmental Monitoring:

- 1) Improved understanding of the environment via data
- 2) Improved Efficiency
- 3) Increased Sustainability
- 4) Business Growth

Basic Steps of Environmental Monitoring using IoT:

- 1) **Observation (Monitor the Environment and Collect Data):** The first step in the environmental monitoring process is to observe and collect data. This involves using sensors or other IoT devices to measure factors such as air quality, temperature and humidity levels. These connected IoT devices gather data about the environment and transmit it to a central hub. From here, the data can be reviewed in real-time or used for further analysis off line.
- 2) **Analysis (Measure Data):** The next step is to analyse the data collected by IoT devices. This includes looking at trends over time, identifying areas of concern and any correlations between environmental variables, time of day, behaviours and the relationships between indoor and outdoor metrics. This data analysis can help businesses measure their environmental footprint and make informed decisions about how to reduce their environmental impact.
- 3) **Storage (Catalogue Data):** Once the data has been analysed, it needs to be stored so that it can be accessed in the future. IoT environmental monitoring systems make this easy by storing the data in a secure cloud-based database, allowing businesses to access the data whenever they need it and analyse how their environmental impact is changing over time.
- 4) **Action (Provide Actionable Insights from the Data and Analysis):** Finally, businesses need to be able to act based on the data that has been gathered and analysed. IoT-enabled environmental monitoring systems can provide insights into how businesses can best reduce their environmental impact, such as by using renewable energy sources or introducing water conservation measures. These actionable insights may involve changing operational processes, implementing new technologies or even making changes to their overall business strategy.

Devices used for Environmental Monitoring:

- 1) **Sensors:** These measure air quality, temperature, humidity, light levels and other factors. They can also be used to detect chemical or water leaks.
- 2) **Data Loggers:** These record and store data over a set period of time. This can be used to measure changes in the environment over time or detect any sudden changes.
- 3) **GIS (Geographic Information System):** This combines mapping technology with real-time data to provide detailed visualisations of environmental conditions.

- 4) **Remote Monitoring Systems:** These systems allow users to monitor environmental conditions remotely and in real-time, providing timely insights into the state of their environment.
- 5) **Drone-based Systems:** Drones can be used to collect aerial data and conduct surveillance of an environment. This helps businesses monitor for potential problems or hazards, such as oil spills or illegal logging.
- 6) **IoT-Enabled Systems:** IoT-enabled systems collect data from multiple sources and provide a comprehensive view of the environment. These systems are used to measure long-term trends, identify areas of concern and monitor environmental changes over time.

3.6 INFRASTRUCTURE MANAGEMENT

Introduction:

Monitoring and controlling operations of urban and rural infrastructures like bridges, railway tracks, on and offshore wind-farms is a key application of the IoT. IoT devices can also be used to control critical infrastructure like bridges to provide access to ships. Usage of IoT devices for monitoring and operating infrastructure is likely to improve incident management and emergency response coordination and quality of service, up-times and reduce costs of operation in all infrastructure related areas.

Infrastructure Planning

Many cities and towns around the world are using IoT solutions to solve various urban problems, such as traffic congestion and to improve the safety and quality of life of their citizens. By placing IoT sensors in various parts of a city and on different kinds of infrastructure, cities can predict future resource needs and ensure that money is invested wisely.

Infrastructure Design

In order to prevent project delays, building firms have begun using Building Information Modelling (BIM). But during the COVID-19 lockdown, BIM has become increasingly important for firms to complete projects on time. With COVID participants unable to physically meet at the site due to travel restrictions, BIM can be used to effectively design 'smart' infrastructure projects. Once a building is complete, data from IoT sensors can be pulled into BIM to help manage operations. Smart buildings can help keep energy usage patterns in check and regulate temperature trends when needed. The data can then be used to optimise future designs.

Infrastructure Sustainability

In order to reduce greenhouse gas emissions, construction field needs to adopt sustainable methods and practices. Sustainable designs do not mean just using eco-friendly, energy-efficient materials or relying on intelligent prefab technology but it involves creating an IoT network and allowing energy companies to drive greater efficiencies.

Infrastructure Construction

For example, cloud-based systems integrate information from the IoT to deliver key insights into projects, these systems allow diverse information to merge into a single broad platform so that decision-makers can manage a project more effectively. With so many processes simultaneously at play, these systems give decision-makers in infrastructure construction companies the power to monitor, track and analyse project performance 24/7. By reducing costs and reducing risks in each project, these systems increase profits for the companies.

Examples of IoT-based Infrastructure Management:

- 1) Smart Grids
- 2) Water Management Systems
- 3) Transportation and Traffic Management
- 4) Asset Tracking and Management
- 5) Building Automation Systems
- 6) Environmental Monitoring
- 7) Waste Management
- 8) Railway and Fleet Management
- 9) Smart City Infrastructure
- 10) Remote Monitoring and Control

3.7 MANUFACTURING

Introduction:

IoT connects consumers, manufacturers and products. The Internet of Things is a global technology that is transforming the industry and manufacturing sector. The IoT wave of Industry takes manufacturing from a purely automated assembly line model of production to a model where the machines are intelligent and communicate with one another. IoT in manufacturing brings with it the opportunity for inserting intelligence into factories. This starts with creating smart objects, which involves embedding sensors, actuators, and controllers into just about everything related to production. Connections tie it together so that people and machines work together to analyze the data and make intelligent decisions. Eventually this leads to machines predicting failures and self-healing and points to a world where human monitoring and intervention are no longer necessary.

Benefits of IoT in Manufacturing:

- 1) IoT recognize manufacturing delays and helps to identify the underlying causes.
- 2) Production units benefit majorly with automation of various processes in the manufacturing industry.
This allows the maximum utilization of raw material and manufacturing components.

- 3) IoT leads to better allocation of resources. It allows users to shift their focus on clients and profits rather than worrying about tedious and time-consuming tasks.

IoT Applications in Manufacturing Industry:

- 1) Intelligent Product Enhancements
- 2) Dynamic Response to Market Demands
- 3) Improved Facility Service
- 4) Product Safety
- 5) Lower Costs, Optimized Resource Use and Waste Reduction
- 6) Quality Control
- 7) Predictive Maintenance
- 8) Inventory Management
- 9) Smart Packaging
- 10) Smart Metering
- 11) Supply Chain Management
- 12) Workshop Monitoring
- 13) Production Flow Monitoring
- 14) Digital Twins

3.8 PHARMACEUTICALS AND HEALTHCARE

Introduction:

In healthcare, IoT-based healthcare systems collect a variety of patient data and get inputs from doctors and medical professionals. All these devices can communicate with each other and take important actions that would provide timely help to save someone's life. The potential application of IoT in healthcare can improve a patient's health, healthcare employee productivity and hospital workflow.

Workflow of IoT Healthcare:

- 1) A sensor collects data from a patient, doctor or nurse inputs data.
- 2) AI-driven algorithms like Machine Learning (ML) are used to analyse the collected data.
- 3) The device decides whether to act or send the information to the cloud.
- 4) Doctors or health practitioners can make actionable and informed decisions based on the data provided by IoT healthcare solutions.

Advantages of IoT in Healthcare:

- 1) Research
- 2) Remote Medical Assistance

- 3) Tracking & Alerts
- 4) Simultaneous Reporting & Monitoring
- 5) Data Assortment & Analysis
- 6) End to End Connectivity & Affordability

Challenges of IoT in Healthcare:

- 1) Data Security & Privacy
- 2) Integrating Multiple Devices
- 3) Data Overload & Accuracy
- 4) Cost

Applications of IoT in Healthcare:

- 1) **Hearables:** Hearables are new-age hearing aids which have completely transformed the way people who suffered hearing loss interact with the world. Nowadays, hearables are compatible with Bluetooth which syncs your smartphone with it. It allows you to filter, equalize and add layered features to real-world sounds.
- 2) **Ingestible Sensors:** They are pill-sized sensors which monitor the medication in our body and warn us if it detects any irregularities in our bodies. These sensors can be a boon for a diabetic patient as would help in reducing/controlling occurrence or severity of symptoms and provide an early warning for critical health issues.
- 3) **Moodables:** Moodables are mood-enhancing devices which help in improving our mood throughout the day. Moodables are head-mounted wearables that send low-intensity currents to the brain which elevates our mood.
- 4) **Healthcare Charting:** IoT devices such as Audemix reduce much manual work which a doctor has to do during patient charting. It is powered by voice commands and captures the patient's data. It makes the patient's data readily accessible for review.
- 5) **Insulin Pens & Smart CGM (Continuous Glucose Monitor):** These devices are used for the real-time monitoring of blood glucose levels and data sharing over a dedicated mobile app. Patients with diabetes can use these devices to track their glucose levels and even send this data to their doctor and the relevant medical staff.
- 6) **Smart Video Pills:** A smart pill travels through a patient's intestinal tract to take its clear-cut picture. It can then send those pictures to a wearable device that is connected with dedicated medical applications. Smart pills are also helpful to visualize the gastrointestinal tract and colon remotely.

IoT Future Predictions in Health Care:

- 1) **Personalized Healthcare:** IoT devices will allow for real-time collection of health data, which can then be used to create tailored treatment plans for individual patients. This will lead to more effective treatment and better outcomes for patients.
- 2) **Remote Patient Monitoring:** With IoT, patients can be monitored remotely, allowing healthcare providers to keep track of their conditions and intervene when necessary. This will result in fewer hospital visits and readmissions, which will ultimately reduce healthcare costs.
- 3) **Predictive Maintenance:** IoT devices will help healthcare providers predict equipment failures and maintenance needs, ensuring that they can provide uninterrupted care to patients.
- 4) **Telemedicine:** Telemedicine services will be enabled by IoT devices, allowing patients to consult with doctors remotely. This will be especially useful for patients who live in remote areas or have mobility problems.

3.9 TRANSPORTATION

Introduction:

Internet of Things (IoT) has crucial applications in the transportation system. IoT plays an important role in all the field of transportation as air-transportation, water-transportation and land transportation. All the component of these transportation fields is built with smart devices (sensors, processors) and interconnected through cloud server or different servers that transmit data to networks.

IoT in transportation is not only for traveling from one place to another, but it also makes safer, greener and more convenient. For example, a smart car performs work simultaneously such as navigation, communication, entertainment, efficient and more reliable travel. IoT facilitates travellers to remain seamlessly connected to every means of travel. The vehicle is connected with the variety of wireless standards to the internet such as Bluetooth, Wi-Fi, 3G, 4G, intelligent traffic system and even to other vehicles. Smart Traffic Management in Barcelona (Spain), Fleet Management with UPS, Smart Parking in San Francisco (USA), Connected Vehicles by Tesla, Smart Public Transportation in Singapore, Cargo Tracking with Maersk Line, etc are some real-world examples of IoT-based Transportation.



Applications of IoT in Transportation:

1) Automated Toll and Ticketing

The traditional tolling and ticketing systems are not only becoming outdated but they are also not proving to be effective for assisting the current flow of vehicles on the road. With the increased number of vehicles on the road, the toll booths have become busy and crowded as well on the highways and the drivers have to spend a lot of time waiting for their turn. The toll booths do not have enough resources and manpower to immediately assist many vehicles. Compared to traditional tolling and ticketing systems, IoT in transportation offers automated tolls. With the help of RFID tags and other smart sensors, managing toll and ticketing have become much easier for traffic police officers.

The majority of advanced vehicles nowadays have IoT connectivity. Any vehicle which might be a kilometre away from the tolling station can easily be detected with the help of IoT technologies. This enables the lifting of the traffic barriers for the vehicles to pass through. However, the older vehicles do not have IoT connectivity, but the smartphones of the car owners can serve the same purpose as well, that is, taking automatic payments through phones linked to the digital wallet. This indicates that IoT in transportation is much more flexible and is compatible with new vehicles and demonstrate easy integration with older vehicles as well, for automated toll and ticketing procedures.

2) Advanced Vehicle Tracking/ Transportation Monitoring

Vehicle tracking or transportation monitoring systems have become the need of many businesses to manage their fleets and supply chain processes effectively. With the help of GPS trackers, transportation companies have smooth access to real-time location, facts and figures about the vehicle. This enables the transportation companies to monitor their important assets in real-time.

Apart from location monitoring, IoT devices can also monitor the driver's behaviour and can inform about the driving style and idling time. In fleet management systems, IoT has minimized the operating and fuel expenditures along with the cost of maintenance. As far as transportation monitoring is concerned, then it

can be said that real-time tracking has made the implementation of smart decisions much easier, enabling the drivers to identify the issues in the vehicle immediately and take precautions where necessary.

3) Smart Traffic Management

Smart Traffic Signals: IoT can help optimize traffic flow by adjusting signal timings based on real-time traffic conditions, reducing congestion and emissions.

Intelligent Traffic Monitoring: Connected sensors and cameras can provide authorities with data on traffic patterns, enabling quicker responses to accidents or traffic disruptions.

4) Vehicle-to-Everything (V2X) Communication

Safety Enhancements: IoT enables vehicles to communicate with each other (V2V) and with infrastructure (V2I) to prevent accidents by sharing data on speed, location, and potential hazards.

Connected Autonomous Vehicles: IoT is critical for self-driving cars, allowing them to gather and share data with other vehicles and systems to make informed driving decisions.

5) Smart Parking Solutions

Parking Availability: IoT-enabled sensors can detect open parking spots and guide drivers to them, reducing the time spent searching for parking and minimizing congestion.

Automated Payment Systems: IoT can enable seamless payment for parking through mobile apps or vehicle-integrated systems.

6) Environmental Monitoring

Emission Control: IoT can monitor vehicle emissions in real-time and alert drivers if their vehicles exceed environmental standards, contributing to cleaner air quality.

Smart Route Planning: IoT can optimize routes to minimize fuel consumption and reduce carbon footprints.

7) Supply Chain Optimization

Asset Tracking: IoT sensors can monitor the location and condition of goods throughout the supply chain, improving transparency and reducing losses.

Cold Chain Monitoring: For temperature-sensitive shipments, IoT can ensure that goods are maintained within optimal temperature ranges, sending alerts if conditions deviate.

Advantages of IoT-based Transportation:

- 1) Reduced Environmental Impact
- 2) Better Customer Experience
- 3) Improved Safety & facilities
- 4) Real Time Monitoring and Tracking
- 5) Optimized Operations and Efficiency
- 6) Supply Chain Visibility and Transparency
- 7) Smart Infrastructure and Urban Planning

UNIT-4

IOT CHALLENGES

4.1 SECURITY & PRIVACY CONCERNS

4.1.1 Security Concerns

1) Lack of Encryption

Although encryption is a great way to prevent hackers from accessing data, it is also one of the leading IoT security challenges. This results in increased attacks where hackers can easily manipulate the algorithms designed for protection. For example, IoT devices frequently fail to encrypt data stored on devices or traveling over the network.

2) Insufficient Testing and Updating

With the increase in the number of IoT devices, IoT manufacturers are eager to produce and deliver their devices as fast as possible without sacrificing security. Most of these devices and IoT products do not receive enough testing and updates and are prone to hackers and other security issues.

3) Brute forcing and the risk of default passwords

Weak credentials and login details leave nearly all IoT devices vulnerable to password hacking and brute force. Any company that uses factory default credentials on their devices is placing both their business and its assets and the customer and their valuable information at risk of being susceptible to a brute force attack.

4) Inadequate Device Security

Inadequate device security refers to the lack of proper measures to protect electronic devices such as computers, smartphones, and IoT devices from cyber-attacks, hacking, data theft, and unauthorized access. This can happen due to outdated software, weak passwords, lack of encryption, and other security risks. It is important to regularly update the software and implement strong security measures to ensure the security and privacy of sensitive information stored on these devices. Many IoT devices have weak security features and can be easily hacked.

5) Lack of Standardization

Lack of standardization refers to the absence of agreed-upon specifications or protocols in a particular field or industry. This can result in different systems, products, or processes being incompatible with each other, leading to confusion, inefficiency, and decreased interoperability. For example, in the context of technology, a lack of standardization can cause difficulties in communication and data exchange between different devices and systems. Establishing standards and protocols can help overcome this and ensure uniformity and compatibility. There is a lack of standardization in IoT devices, making it difficult to secure them consistently.

6) Vulnerability to Network Attacks

Vulnerability to network attacks refers to the susceptibility of a network, system, or device to being compromised or exploited by cybercriminals. This can happen due to weaknesses in the network infrastructure, unpatched software, poor password management, or a lack of appropriate security measures.

Network attacks can result in data theft, loss of privacy, disruption of services, and financial loss. To reduce vulnerability to network attacks, it's important to implement strong security measures such as firewalls, encryption, and regular software updates, as well as educate users on safe internet practices. IoT devices rely on networks, making them vulnerable to attacks like denial-of-service (DoS) attacks.

7) Unsecured Data Transmission

Unsecured data transmission refers to the transfer of data over a network or the internet without adequate protection. This can leave the data vulnerable to interception, tampering, or theft by malicious attackers. Unsecured data transmission can occur when data is transmitted over an unencrypted network connection or when insecure protocols are used. To protect sensitive data during transmission, it is important to use secure protocols such as SSL/TLS or VPN and to encrypt the data before sending it. This can help to ensure the confidentiality and integrity of the data, even if it is intercepted during transmission. IoT devices often transmit sensitive data, which may be vulnerable to eavesdropping or tampering if not properly secured.

8) Insider Threats

Insider threats refer to security risks that come from within an organization, rather than from external sources such as hackers or cyber criminals. These threats can take many forms, such as employees who intentionally or unintentionally cause harm to the organization, contractors who misuse their access privileges, or insiders who are coerced into compromising the security of the organization. Insider threats can result in data breaches, theft of intellectual property, and damage to the reputation of the organization. To mitigate the risk of insider threats, organizations should implement strict access controls, monitor employee activity, and provide regular training on security and privacy policies. Additionally, organizations should have a plan in place to detect, respond to, and recover from security incidents involving insiders. Employees or contractors with access to IoT systems can pose a security risk if they intentionally or unintentionally cause harm.

4.1.2 Privacy Concerns

Privacy concerns refer to issues related to the collection, storage, use, and sharing of personal information. This can include concerns about who has access to personal information, how it is being used, and whether it is being protected from unauthorized access or misuse. In the digital age, privacy concerns have become increasingly important as personal information is being collected and stored on an unprecedented scale. To address privacy concerns, individuals and organizations need to implement appropriate security measures to protect personal information, be transparent about how it is being used, and respect individuals' rights to control their information. Additionally, privacy laws and regulations have been established to provide guidelines and protections for individuals' personal information. The vast amount of data generated by IoT devices raises privacy concerns, as personal information could be collected and used without consent.

To address these challenges, it is important to implement security measures such as encryption, secure authentication, and software updates to ensure the safe and secure operation of IoT devices and systems.

4.2 FRAGMENTATION OF STANDARDS

The term "fragmentation" in the context of IoT (Internet of Things) typically refers to the challenges and issues arising from the lack of standardization in various aspects of IoT technologies. Standardization is crucial for ensuring interoperability, security, and seamless integration of diverse IoT devices and systems. When there is fragmentation in the IoT landscape, it can lead to several problems as mentioned below:

1) Interoperability Issues

IoT devices and platforms may use different communication protocols, data formats, or standards. This can create interoperability challenges, making it difficult for devices from different manufacturers to work together seamlessly.

2) Security Concerns

In a fragmented IoT ecosystem, there may be variations in security implementations. Some devices might have robust security features, while others may lack adequate protection. This inconsistency can create vulnerabilities and increase the risk of security breaches.

3) Complex Development and Integration

Developers face challenges when creating applications that need to work across a diverse range of IoT devices. The lack of standardized interfaces and communication protocols increases the complexity of software development and integration efforts.

4) Increased Costs

Manufacturers and developers may incur higher costs due to the need to customize solutions for different standards. Additionally, users may need to invest in specific infrastructure or middleware to bridge the gaps between fragmented systems.

5) Limited Innovation

Fragmentation can stifle innovation as developers and companies may be hesitant to invest resources in creating new solutions if they are uncertain about the compatibility and widespread adoption of those solutions.

4.3 SCALABILITY

Scalability is an important consideration when it comes to IoT (Internet of Things) systems, as these systems typically involve a large number of devices and can generate massive amounts of data. The following are some key factors that can impact the scalability of IoT systems:

1) Network Capacity

As the number of devices and data traffic increases, it is important to have enough network capacity to handle the increased demand. This may require upgrading network infrastructure, such as adding more access points, deploying mesh networks, or increasing bandwidth.

2) Data Processing

As the amount of data generated by IoT devices increases, it can become challenging to process and analyze that data in a timely and efficient manner. One approach is to use edge computing, which enables data processing to occur closer to the source, reducing latency and network traffic. Cloud computing can also be used to process large volumes of data.

3) Device Management

With large numbers of devices, it can become difficult to manage and maintain them all. This requires a scalable device management system, which may include features such as automated provisioning, remote management, and over-the-air updates.

4) Security

As the number of devices and data traffic increases, security becomes an increasingly important consideration. A scalable security system should include features such as encryption, authentication, and access control and should be able to adapt to changing threats and vulnerabilities.

5) Interoperability

IoT devices are often produced by different manufacturers and use different protocols, which can make integration and interoperability challenging. A scalable IoT system should support interoperability between different devices and protocols, allowing for more efficient and effective data exchange.

To address scalability issues in IoT, it's important to design systems with scalability in mind from the beginning. This includes using scalable architectures, adopting distributed computing paradigms, implementing efficient data management strategies, and leveraging technologies such as cloud computing and edge computing. Additionally, ongoing monitoring, optimization, and the use of scalable protocols can contribute to a more scalable and robust IoT ecosystem.

4.4 COMPATIBILITY BETWEEN DIFFERENT SMART SENSORS

4.4.1 Temperature Sensors

In the Internet of Things (IoT) ecosystem, temperature sensors play a crucial role in collecting data for smart and automated systems. These sensors enable devices to monitor temperature changes and share this data over the Internet, allowing real-time analysis and decision-making. IoT temperature sensors are integral to various applications, including smart homes, industrial automation, healthcare, agriculture, environmental monitoring, etc.

Applications:

1) Smart Homes

Climate Control: IoT-enabled temperature sensors can help optimize HVAC systems for energy-efficient heating and cooling based on real-time temperature readings.

Safety: Used in smoke and fire detectors to trigger alarms and notifications when temperature thresholds are surpassed.

Smart Appliances: Integration in devices like refrigerators, ovens, and water heaters for temperature regulation and remote control.

2) Industrial Automation

Predictive Maintenance: Monitoring machinery and equipment temperatures to predict failures or schedule maintenance before issues arise.

Process Monitoring: Ensuring consistent temperature conditions in manufacturing processes to maintain product quality.

Supply Chain Management: Temperature sensors in logistics to monitor and control conditions during the transport of temperature-sensitive goods.

3) Healthcare

Patient Monitoring: Wearable devices equipped with temperature sensors for real-time tracking of body temperature and early detection of fevers or health anomalies.

Pharmaceutical Storage: Ensuring proper storage conditions for medicines and vaccines by monitoring and maintaining required temperature ranges.

4) Agriculture

Greenhouse Management: IoT-enabled temperature sensors to automate and optimize greenhouse environments for optimal plant growth.

Soil Temperature Monitoring: Helps in determining the right conditions for planting and improving crop yield.

Cold Chain for Fresh Produce: Monitoring the temperature of storage units to prevent spoilage during transport and storage.

5) Energy Management

Smart Grids: Temperature data assists in regulating grid performance and detecting potential overheating or failures.

Renewable Energy Systems: Monitoring the temperature of solar panels or wind turbines to optimize performance and prevent damage.

6) Smart Cities

Environmental Monitoring: Used in urban areas to monitor environmental conditions and collect data for climate and pollution studies.

Building Management Systems: Integrated in commercial buildings for efficient temperature control, reducing energy consumption, and enhancing comfort.

7) Food and Beverage Industry

Cold Storage Monitoring: Ensuring food products are stored at safe temperatures during processing, storage, and transport.

Quality Control: Maintaining optimal temperatures for food preparation and safety in food service establishments.

8) Automotive and Transportation

Vehicle Monitoring: Sensors can detect engine temperature and cabin temperature to provide feedback for performance and passenger comfort.

Cargo Temperature Monitoring: Ensuring perishable goods are transported at appropriate temperatures with real-time alerts for any deviations.

9) Research and Development

Scientific Experiments: IoT temperature sensors are used in laboratories for precise monitoring of temperature-sensitive experiments.

Data Collection for Analytics: Large-scale temperature data collection for various research applications, such as climate modeling and product testing.

10) Disaster Management

Wildfire Detection: Sensors can be placed in forests to detect sudden temperature spikes and alert authorities to potential fire hazards.

Weather Monitoring Systems: Integrated into broader IoT weather monitoring networks for real-time temperature data collection and forecasting.

Advantages:

1) Real-time Monitoring

Temperature sensors enable continuous and real-time monitoring of environmental conditions, which is essential for responsive IoT applications like climate control and supply chain management.

2) Automation and Control

They can trigger automated actions based on temperature readings, such as turning on air conditioning, adjusting thermostat settings, or sending alerts in case of overheating.

3) Energy Efficiency

IoT systems that utilize temperature sensors can optimize energy usage by dynamically adjusting heating and cooling based on current needs, leading to reduced energy consumption and lower costs.

4) Improved Safety

In industrial settings, temperature sensors can detect potential issues such as overheating machinery, helping to prevent equipment damage and reduce the risk of fire hazards.

5) Data Collection and Analysis

Sensors provide valuable data that can be used for predictive analytics, trend analysis, and forecasting, which can improve decision-making and system performance over time.

6) Versatile Applications

Temperature sensors can be used in a wide range of environments and industries, from healthcare (monitoring patient temperatures) to agriculture (monitoring soil and climate conditions).

Disadvantages:

1) Connectivity Issues

IoT devices rely on stable internet connectivity to transmit data. If there is a network failure, real-time monitoring and remote access could be compromised.

2) Security and Privacy Concerns

IoT temperature sensors, like all connected devices, are vulnerable to cyber-attacks. Unauthorized access to sensor data could lead to privacy issues or tampering with environmental controls.

3) Power Consumption

Continuous operation of temperature sensors, especially in remote or battery-operated IoT devices, can lead to significant power consumption, requiring regular maintenance or energy-efficient solutions.

4) Calibration and Accuracy

Sensors may require regular calibration to maintain accuracy. Incorrect readings can lead to improper automation responses, affecting system reliability.

5) Environmental Limitations

Some temperature sensors may not function effectively in extreme conditions (e.g., very high or very low temperatures), which can limit their applicability in certain IoT scenarios.

6) Cost of Implementation

Deploying a large number of temperature sensors, along with the necessary networking infrastructure and maintenance, can be costly, especially for large-scale projects.

7) Integration Challenges

Integrating temperature sensors with existing IoT ecosystems and ensuring interoperability with other devices and platforms can be complex and may require custom solutions.

Factors to Consider for Compatibility of Temperature Sensor

The compatibility of temperature sensors depends on various factors, including the type of sensor, its intended application, and the system it needs to interface with. The following are the key compatibility considerations for temperature sensors:

1) Interface and Communication

Analog Output: Many temperature sensors provide analog voltage or current output (e.g., thermocouples, thermistors), which needs to be converted to digital signals using an analog-to-digital converter (ADC).

Digital Output: Sensors like semiconductor temperature sensors can provide digital data via communication protocols such as I2C, SPI, or 1-Wire, making them compatible with modern microcontrollers and processors.

Special Interfaces: Some advanced sensors have proprietary interfaces that require specific hardware or software support for full compatibility.

2) Power Requirements

Low-Power Applications: Certain temperature sensors, especially semiconductor-based ones, are designed for low-power operation, making them suitable for battery-powered devices.

High-Power Considerations: Some sensors, like IR sensors with advanced optics, may require more power and therefore need to be paired with systems that can supply adequate power.

3) Environmental Compatibility

Temperature Range: Ensure the sensor is compatible with the operating temperature range of your application. For example, thermocouples can withstand high temperatures, while certain semiconductor sensors may have a more limited range.

Durability: In industrial or harsh environments, compatibility with protective enclosures or sensor housings is important to prevent damage from exposure to moisture, dust, or chemicals.

4) Signal Conditioning

Amplifiers and Filters: Some sensors, like thermocouples and RTDs, require signal conditioning (amplification, filtering, etc.) for accurate readings, which must be compatible with the existing system.

Cold Junction Compensation: Thermocouples require cold junction compensation to provide accurate temperature readings, so compatibility with the system's compensation method is necessary.

5) Calibration and Accuracy Requirements

Calibration Needs: RTDs and certain thermistors may need periodic calibration to maintain accuracy, and the system they integrate with must support this process.

Accuracy Requirements: Different applications demand varying levels of precision. For example, medical devices may require high-accuracy sensors compatible with stringent calibration protocols, while HVAC systems may use less precise sensors.

6) Software Compatibility

Drivers and Libraries: Sensors that communicate digitally may need compatible drivers or libraries for seamless integration with microcontrollers, computers, or IoT platforms.

Data Formats: Ensure that the data output format from the sensor matches the input requirements of the data processing system.

4.4.2 Humidity Sensor

Humidity sensors are devices designed to measure and quantify the moisture content or humidity level in the air or other gases. Humidity sensors play a crucial role in the Internet of Things (IoT) by enabling the measurement and monitoring of humidity levels in various environments. These sensors are integral components in IoT systems designed for applications such as smart homes, agriculture, industrial processes, healthcare, and environmental monitoring. The ability to accurately measure humidity is essential for ensuring optimal conditions in diverse settings.

Applications:

1) Smart Agriculture

Precision Farming: Humidity sensors help monitor soil moisture levels and atmospheric humidity, enabling more precise irrigation systems. This leads to better water management, improved crop yields, and reduced water waste.

Greenhouse Automation: IoT-connected humidity sensors ensure optimal conditions for plant growth by controlling ventilation, watering systems, and temperature automatically.

2) Smart Homes

HVAC Systems: Integrated with IoT, humidity sensors can help regulate home heating, ventilation, and air conditioning systems to maintain comfort and energy efficiency.

Mold Prevention: Humidity sensors can detect high moisture levels that could lead to mold growth and send alerts or trigger dehumidifiers automatically.

3) Weather Monitoring

Environmental Stations: IoT-enabled humidity sensors are essential for collecting real-time data in weather stations for accurate weather forecasting and climate monitoring.

Disaster Preparedness: These sensors play a crucial role in detecting humidity changes that may indicate natural phenomena like thunderstorms or heavy rain.

4) Healthcare Monitoring

Indoor Air Quality Control: In hospitals and clinics, maintaining proper humidity is critical for patient comfort and health. IoT-connected sensors help manage air quality and prevent airborne transmission of diseases.

Smart Medical Devices: Some medical devices use humidity sensors to monitor the environment and ensure optimal operating conditions for both equipment and patient care.

5) Industrial Applications

Storage and Warehousing: Humidity sensors ensure that warehouses storing sensitive products, such as food, pharmaceuticals, and electronics, maintain proper humidity levels to prevent spoilage or damage.

Manufacturing Processes: Certain manufacturing processes require controlled humidity levels to maintain product quality. IoT-enabled sensors can automate and monitor these conditions efficiently.

6) Smart Cities

Public Transportation Systems: Monitoring humidity in subway systems, train stations, and other public transit areas helps maintain air quality and passenger comfort.

Building Management Systems (BMS): Humidity sensors integrated into IoT systems help optimize energy usage in large buildings by controlling HVAC and ventilation systems.

7) Supply Chain and Cold Chain Logistics

Perishable Goods Monitoring: IoT humidity sensors track environmental conditions during the transport of perishable goods, ensuring that they remain within optimal humidity levels and reducing the risk of spoilage.

Alert Systems: Sensors can send real-time alerts to logistics teams if humidity levels go out of the safe range, enabling quick responses.

8) Museums and Art Preservation

Environmental Control: IoT-connected humidity sensors help maintain strict humidity levels in museums, galleries, and archives to preserve delicate artworks and historical documents.

9) Smart Offices

Employee Comfort and Productivity: IoT systems using humidity sensors can regulate indoor climate conditions, improving comfort and enhancing productivity.

Energy Efficiency: Automatic adjustments of air conditioning and heating based on real-time humidity data help in reducing energy consumption.

10) Agri-food Sector

Food Processing Plants: Humidity sensors help maintain optimal conditions in processing facilities, ensuring that food products remain safe and high-quality.

Drying and Curing Processes: These sensors aid in controlling the drying and curing stages in food production by adjusting to precise humidity requirements.

Advantages:

1) Environmental Monitoring

Humidity sensors are essential for IoT systems used in environmental monitoring, agriculture, and weather forecasting. They enable precise data collection on humidity levels, which is crucial for maintaining optimal conditions for crops, greenhouses, and indoor spaces.

2) Energy Efficiency

In smart homes and industrial settings, humidity sensors can be part of HVAC (Heating, Ventilation, and Air Conditioning) systems to optimize energy usage based on ambient humidity, contributing to reduced energy consumption.

3) Health and Comfort

IoT applications using humidity sensors help maintain appropriate humidity levels, which can prevent health issues related to overly dry or humid air. This is particularly beneficial in homes, offices, and hospitals.

4) Data-Driven Decisions

In combination with IoT platforms, humidity sensors provide real-time data and historical trends that can be used for predictive maintenance and decision-making processes in various industries.

5) Integration with Smart Systems

Humidity sensors can be easily integrated with IoT devices and networks to automate responses, such as turning on dehumidifiers or adjusting ventilation systems.

Disadvantages:

1) Maintenance and Calibration

Humidity sensors require regular calibration and maintenance to ensure accurate readings over time. In remote IoT applications, this can be challenging and costly.

2) Sensitivity to Environment

Humidity sensors can be affected by contaminants, dust, and condensation, potentially leading to inaccurate data or sensor failure.

3) Power Consumption

While many IoT sensors are designed to be energy-efficient, humidity sensors integrated into continuous monitoring systems can contribute to power drain, especially in battery-powered IoT devices.

4) Initial Cost

High-quality humidity sensors can be expensive, which may be a barrier for large-scale IoT implementations where numerous sensors are required.

5) Data Security and Privacy

As with any IoT device, humidity sensors can pose potential security risks if not properly protected. Data breaches or unauthorized access to sensor data can compromise the integrity of IoT systems.

6) Limited Measurement Range

Some humidity sensors may not be effective in extreme environmental conditions, impacting their performance and suitability for certain IoT applications.

Factors to Consider for Compatibility of Humidity Sensor:

1) Output Signal:

Different humidity sensors may have analog or digital output signals. Ensure compatibility with the data acquisition or control system.

2) Accuracy Requirements:

Consider the required accuracy for your application and choose a sensor that meets those requirements. Calibration may be necessary for optimal accuracy.

3) Environmental Conditions:

Verify that the humidity sensor is suitable for the environmental conditions of the application, including temperature, contaminants, and potential exposure to liquids.

4) Response Time:

Consider the required response time for humidity changes in your application. Some sensors may respond more quickly than others.

5) Interfacing:

Ensure that the sensor can be easily interfaced with your microcontroller or data acquisition system. Check for available libraries or protocols.

6) Calibration:

Some humidity sensors may require regular calibration to maintain accuracy. Understand the calibration process and frequency.

7) Power Consumption:

Consider the power requirements of the humidity sensor, especially in low-power or battery-operated applications.

8) Application-Specific Considerations:

Different applications (e.g., industrial, HVAC, consumer electronics) may have specific requirements. Choose a sensor that aligns with those considerations.

4.4.3 Proximity Sensor

A proximity sensor is an essential type of sensor that detects the presence or absence of an object or its distance from the sensor without making physical contact. In the context of IoT (Internet of Things), proximity sensors enable devices to "sense" their environment and interact with it in real time, providing data that can be used for automation, monitoring, and control of various systems.

IoT systems typically connect proximity sensors to other devices or platforms over networks (e.g., Wi-Fi, Bluetooth, Zigbee, etc.), allowing remote monitoring and control. The sensor's data can be integrated with cloud platforms for analysis, decision-making, and triggering automated responses, creating smarter and more efficient systems.

Applications:

1) Smart Homes

Lighting Control: Proximity sensors can detect the presence of people in a room and automatically turn lights on or off, contributing to energy savings and convenience.

Security Systems: Used in motion detectors to identify unauthorized access or activity, sending real-time alerts to homeowners or security teams.

Appliance Automation: Sensors can trigger appliances like fans or air conditioners when someone enters a room, improving comfort and efficiency.

2) Industrial Automation

Machine Monitoring: Proximity sensors help in detecting the position of components, ensuring smooth operation and preventing malfunction in industrial machinery.

Safety Systems: Used for ensuring safety by detecting the presence of personnel in restricted areas or detecting obstacles in robotic systems.

Inventory Management: Proximity sensors can help track the presence or movement of goods on assembly lines, automating inventory counts and minimizing human errors.

3) Automotive Applications

Parking Assistance: In vehicles, proximity sensors help with parking by detecting objects around the vehicle and providing feedback to the driver.

Collision Detection: Proximity sensors are part of advanced driver assistance systems (ADAS) that alert drivers to nearby vehicles, pedestrians, or obstacles to avoid accidents.

4) Healthcare

Patient Monitoring: Proximity sensors in healthcare settings can track patient movement, monitor room entry/exit, or detect the approach of medical staff, enhancing patient care.

Touchless Interfaces: In hospitals, sensors enable touchless operation of equipment (e.g., sinks, doors), reducing the risk of contamination.

5) Smart Cities

Smart Traffic Systems: Proximity sensors in roads or traffic lights help manage traffic flow by detecting the presence of vehicles or pedestrians.

Public Transport: Sensors on buses or trains can detect occupancy, optimize routes, or improve maintenance scheduling based on passenger proximity and usage patterns.

6) Retail and Shopping

Customer Interaction: Proximity sensors can track when a customer approaches a display or product, triggering targeted advertisements or notifications on their smartphone.

Inventory Management: Used in automated systems to track stock levels and alert store managers when products need restocking or when items are out of place.

7) Robotics

Obstacle Detection and Avoidance: Proximity sensors are key for robots to sense obstacles in their path, helping them navigate environments autonomously.

Human-Robot Interaction: In collaborative robots (cobots), proximity sensors help detect human presence to ensure safety and prevent accidents.

Advantages:

1) Automation and Efficiency

Hands-Free Operation: Proximity sensors allow for automated, hands-free operation in many IoT systems. For example, they can trigger lights or doors to open when a person approaches, which is common in smart homes or offices.

Energy Efficiency: By detecting the presence of people or objects, proximity sensors help in energy-saving systems. For example, lighting or HVAC systems can automatically turn off when a room is empty, reducing energy waste.

2) Improved Safety

Collision Avoidance: In industrial environments, proximity sensors help ensure safety by detecting objects or personnel near machines or moving parts, thus preventing accidents.

Security: In smart security systems, proximity sensors are used for detecting unauthorized access or monitoring the movement of people or objects around restricted areas.

3) Enhanced User Experience

Intuitive Interfaces: In smart homes or consumer devices, proximity sensors can enhance user experience by making devices more intuitive. For instance, sensors can activate a device when a user approaches and deactivate it when they move away.

Touchless Interaction: These sensors enable touchless interaction with devices, which has become especially valuable in health-conscious environments (e.g., public spaces, and hospitals).

4) Integration with Other IoT Devices

Seamless Integration: Proximity sensors can easily integrate with other IoT systems to create interconnected solutions. For example, they can trigger events in a larger IoT ecosystem, such as turning on a smart thermostat when someone enters a room.

Remote Monitoring and Control: IoT-enabled proximity sensors allow remote monitoring of environments, such as tracking inventory in a warehouse or detecting occupancy in a building.

5) Cost-Effective

Low-Cost Sensors: Many proximity sensors (such as capacitive or inductive sensors) are relatively inexpensive, making them a cost-effective option for large-scale IoT implementations.

Disadvantages:

1) Limited Range

Short Detection Range: Many proximity sensors have a limited detection range (often a few centimetres to a couple of meters). This may not be suitable for applications where a longer detection range is required.

Environment Sensitivity: The range and reliability of proximity sensors can be influenced by environmental factors, such as temperature, humidity, or the presence of certain materials that interfere with the sensor's detection capability.

2) False Triggering

Environmental Interference: Proximity sensors, especially infrared (IR) or ultrasonic types, can be prone to interference from other environmental factors, such as reflective surfaces, heat sources, or external light sources. This can lead to false readings or triggering events at undesired times.

Detection of Unwanted Objects: In some cases, proximity sensors might trigger actions due to objects or people that are not part of the intended detection range, leading to operational inefficiencies or false alarms.

3) Limited to Specific Object Types

Material Sensitivity: Different types of proximity sensors may only detect certain types of materials. For example, inductive sensors detect metallic objects, while capacitive sensors work best with non-metallic objects. This can limit the versatility of sensors in some applications.

Size of Object: Some proximity sensors may struggle to detect very small objects or objects with irregular shapes, which can limit their usability in specific IoT scenarios.

4) Power Consumption (for certain types)

Power Requirements: While many proximity sensors are low-power, certain advanced models (such as ultrasonic or radar-based sensors) may consume more power. This can be a disadvantage for IoT devices that rely on battery power and need to function for extended periods.

IoT Network Overhead: In large-scale IoT systems, proximity sensors continuously sending data to a central system could contribute to network congestion and increased power consumption, especially in wireless IoT applications.

5) Calibration and Maintenance

Need for Calibration: Proximity sensors may require periodic calibration to maintain accuracy, especially in industrial or challenging environments where conditions change over time.

Sensitivity to Wear and Tear: In harsh environments, sensors (particularly mechanical types) can degrade over time, requiring maintenance or replacement to ensure continued performance.

6) Security and Privacy Concerns

Data Privacy: Proximity sensors in consumer IoT devices, like smart doorbells or security systems, can raise privacy concerns. Since they often detect personal movements or locations, there's a risk of data being misused if not securely handled.

Hacking Risks: Proximity sensors that are part of a connected IoT system may become targets for cyberattacks if the system is not properly secured, potentially allowing unauthorized access or manipulation.

7) Limited Advanced Functionality

Basic Detection: While proximity sensors can detect the presence of objects, they often do not provide detailed information such as the object's size, shape, or orientation. In more advanced IoT applications requiring detailed data, additional sensors (e.g., cameras or advanced motion sensors) may be needed.

Factors to Consider for Compatibility of Proximity Sensor:

1) Detection Range

Range Specifications: Consider the distance over which the sensor can detect objects. Different types of sensors have different effective ranges. For example, inductive sensors usually have a shorter range, while ultrasonic sensors can detect objects at greater distances.

Application Needs: Match the sensor's range with the specific requirements of your IoT application to ensure it functions effectively within the environment.

2) Environmental Conditions

Temperature and Humidity: Ensure the sensor is rated for the temperature and humidity conditions of the deployment environment. Some sensors may be less effective in extreme conditions.

Dust, Water, and Chemicals: If the sensor will be used in a harsh environment, look for models with protective ratings (e.g., IP65 or higher) that offer resistance to dust, water, and chemical exposure.

Lighting and Reflective Surfaces: IR sensors, in particular, can be affected by bright light or reflective surfaces, leading to false detections.

3) Power Requirements

Voltage and Current Compatibility: Verify that the sensor's power requirements match the power supply available in your IoT system. Low-power sensors are preferable for battery-operated devices.

Energy Efficiency: Consider sensors that are optimized for low power consumption, especially if the system is intended for continuous or long-term operation.

4) Interface and Communication Protocol

Output Type: Check if the sensor outputs digital, analog, or other signal types. Ensure the output is compatible with the IoT device or microcontroller in use.

Communication Compatibility: For sensors used in IoT applications, it's essential to select models that can easily interface with the system's communication protocols, such as I2C, SPI, UART, or wireless options like Bluetooth and Wi-Fi.

Integration with Existing Systems: Make sure the sensor integrates smoothly with the existing hardware and software platforms.

5) Accuracy and Reliability

Detection Precision: Assess the sensor's ability to provide consistent and accurate readings. Some sensors may have better resolution and response times than others.

Error Tolerance: In applications where precision is critical, choose sensors with low error rates and high reliability.

Durability and Lifespan: Consider how long the sensor is expected to function without significant performance degradation. For IoT applications involving constant use, selecting a robust and reliable sensor is crucial to minimize maintenance and replacements.

UNIT-5

INTRODUCTION TO ARDUINO & RASPBERRY PI

5.1 ARCHITECTURE, PROGRAMMING & APPLICATIONS OF ARDUINO

Introduction:

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

Arduino was born at the Ivrea Interaction Design Institute as an easy tool for fast prototyping, aimed at students without a background in electronics and programming. As soon as it reached a wider community, the Arduino board started changing to adapt to new needs and challenges, differentiating its offer from simple 8-bit boards to products for IoT applications, wearable, 3D printing, and embedded environments.

Advantages of Arduino:

- **Inexpensive:** Arduino boards are relatively inexpensive compared to other microcontroller platforms. The least expensive version of the Arduino module can be assembled by hand, and even the pre-assembled Arduino modules cost less than \$50.
- **Cross-platform:** The Arduino Software (IDE) runs on Windows, Macintosh OSX, and Linux operating systems. Most microcontroller systems are limited to Windows.
- **Simple, clear programming environment:** The Arduino Software (IDE) is easy-to-use for beginners, yet flexible enough for advanced users to take advantage of as well. For teachers, it's conveniently based on the Processing programming environment, so students learning to program in that environment will be familiar with how the Arduino IDE works.
- **Open source and extensible software:** The Arduino software is published as open source tools, available for extension by experienced programmers. The language can be expanded through C++ libraries, and people wanting to understand the technical details can make the leap from Arduino to the AVR C programming language on which it's based. Similarly, you can add AVR-C code directly into your Arduino programs if you want to.
- **Open source and extensible hardware:** The plans of the Arduino boards are published under a Creative Commons license, so experienced circuit designers can make their own version of the module, extending it and improving it. Even relatively inexperienced users can build the breadboard version of the module in order to understand how it works and save money.

Types of Arduino:

1. Arduino Uno

- **Microcontroller:** ATmega328P
- **Features:** 14 digital I/O pins (6 PWM), 6 analog inputs, USB connection, power jack.
- **Best for:** Beginners and educational purposes.

2. Arduino Mega 2560

- **Microcontroller:** ATmega2560
- **Features:** 54 digital I/O pins (15 PWM), 16 analog inputs, 4 UARTs (hardware serial ports).
- **Best for:** Projects requiring more I/O pins, such as large-scale sensor arrays.

3. Arduino Nano

- **Microcontroller:** ATmega328P (similar to Uno)
- **Features:** Compact size, 14 digital I/O pins, 8 analog inputs, USB connection (Mini USB or Micro USB).
- **Best for:** Projects where space is limited.

4. Arduino Leonardo

- **Microcontroller:** ATmega32u4
- **Features:** Built-in USB communication, can simulate a mouse or keyboard.
- **Best for:** Projects requiring USB HID capabilities.

5. Arduino Micro

- **Microcontroller:** ATmega32u4 (similar to Leonardo)
- **Features:** Smaller than the Leonardo, 20 digital I/O pins, USB communication.
- **Best for:** Embedded projects needing direct USB communication.

6. Arduino Due

- **Microcontroller:** Atmel SAM3X8E ARM Cortex-M3
- **Features:** 32-bit ARM core, 54 digital I/O pins, 12 analog inputs, faster processing.
- **Best for:** More complex projects needing higher processing power.

7. Arduino Pro Mini

- **Microcontroller:** ATmega328P
- **Features:** Minimalistic design without USB connection (requires an external FTDI adapter for programming), 14 digital I/O pins.
- **Best for:** Small, portable projects.

8. Arduino Nano Every

- **Microcontroller:** ATmega4809
- **Features:** Enhanced version of Arduino Nano with more capabilities.
- **Best for:** More advanced applications than the original Nano.

9. Arduino MKR Series

- **Microcontroller:** Varies (e.g., SAMD21 Cortex-M0+)
- **Features:** Compact boards with built-in connectivity options such as Wi-Fi, LoRa, and GSM.
- **Best for:** IoT and connectivity-focused projects.

10. Arduino Yún

- **Microcontroller:** ATmega32u4 + Atheros AR9331 (Linux-based system)
- **Features:** Integrated Wi-Fi and Ethernet, Linux capabilities.
- **Best for:** IoT projects needing both microcontroller functionality and Linux capabilities.

11. Arduino Nano 33 IoT

- **Microcontroller:** SAMD21 Cortex-M0+ and Wi-Fi module.
- **Features:** Built-in Wi-Fi and Bluetooth.
- **Best for:** IoT projects requiring wireless connectivity.

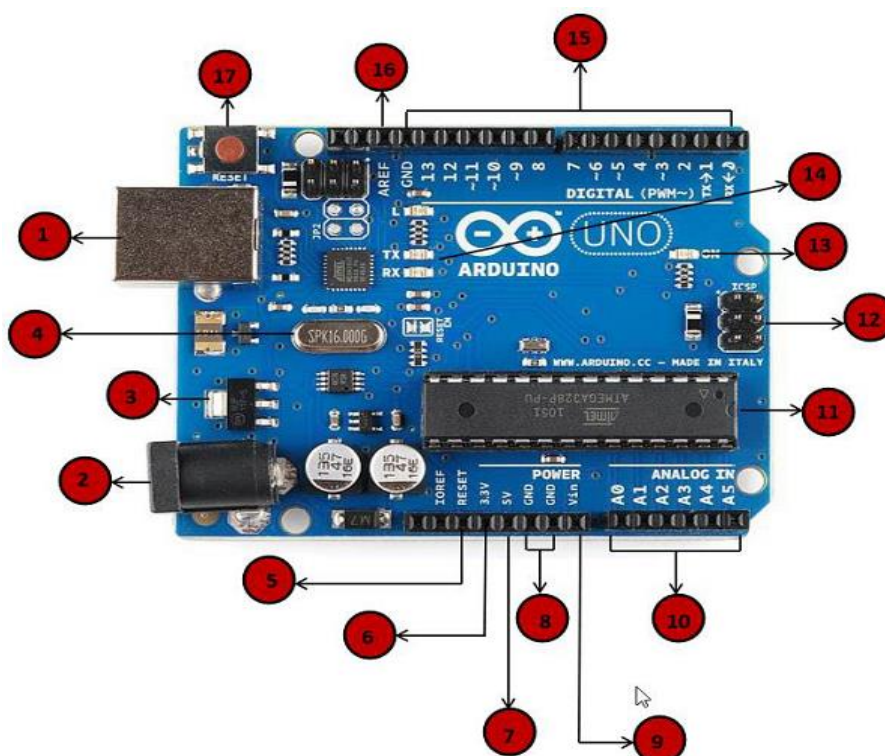
12. Arduino LilyPad

- **Microcontroller:** ATmega328P
- **Features:** Designed for wearables, has sewable pads for easy integration with fabric.
- **Best for:** Wearable tech and e-textiles.

13. Arduino Zero

- **Microcontroller:** SAMD21 Cortex-M0+
- **Features:** 32-bit core, more power than typical 8-bit Arduinos.
- **Best for:** Projects needing better precision and processing power.

5.1.1 Architecture of Arduino



As shown in the above figure, labelled components of the Arduino board are explained as follows:

- 1) **USB:** Arduino board can be powered by using the USB cable from your computer. All you need to do is connect the USB cable to the USB connection. It can be used for both power and communication with the IDE.
- 2) **Barrel Jack:** It is used for power supply. Arduino boards can be powered directly from the AC mains power supply by connecting it to the Barrel Jack.
- 3) **Voltage Regulator:** It regulates and stabilizes the input and output voltages. The function of the voltage regulator is to control the voltage given to the Arduino board and stabilize the DC voltages used by the processor and other elements.
- 4) **Crystal Oscillator:** It keeps track of time and regulates processor frequency. The crystal oscillator helps Arduino in dealing with time issues. How does Arduino calculate time? The answer is, by using the crystal oscillator. The number printed on top of the Arduino crystal is 16.000H9H. It tells us that the frequency is 16 MHz.
- 5) **Reset Pin:** It can be used to reset the Arduino Uno.
- 6) **3.3V Pin:** It can be used as a 3.3V output.
- 7) **5V Pin:** It can be used as a 5V output.
- 8) **GND Pin:** It can be used to ground the circuit.
- 9) **Vin Pin:** It can be used to supply power to the board.
- 10) **Analog Pins (A0-A5):** It can be used to read analog signals to the board.
- 11) **Microcontroller (ATMega328):** The processing and logical unit of the board.
- 12) **ICSP Pin:** A programming header on the board also called SPI.
- 13) **Power Indicator LED:** It indicates the power status of the board.
- 14) **RX and TX LEDs:** It receive (RX) and transmit (TX) LEDs, blink when sending or receiving serial data respectively.
- 15) **Digital I/O Pins:** 14 pins capable of reading and outputting digital signals; 6 of these pins are also capable of PWM.
- 16) **AREF Pins:** It can be used to set an external reference voltage as the upper limit for the analog pins.
- 17) **Reset button:** It can be used to reset the board.

5.1.2 Programming of Arduino

1. Download & install the Arduino environment (IDE)

If you just got your Arduino Uno board, you'll first have to install the Arduino IDE (Integrated Development Environment) on another computer. The code is typed into the IDE and sent to the Arduino via a USB cable. Visit [arduino.cc](https://www.arduino.cc) to download the most recent Arduino IDE version for your computer. There are different versions for Mac, Windows and Linux OS.

- At the download page, click on the “Installer” option for the easiest installation then

- Save the .exe file to your disk drive.
- Open the .exe file.
- Click the button to agree to the licensing agreement.
- Decide which components to put in, then click “Next”.
- Select which folder to put in the program to, then click “Install”.
- Wait for the program to complete installing, then click “Close”.

2. Launch the Arduino IDE

After your Arduino IDE software is downloaded, unzip the folder. To do so, double-click on the Arduino shortcut on your Desktop. The IDE will open up and you’ll see the code editor.

3. If needed, install the drivers

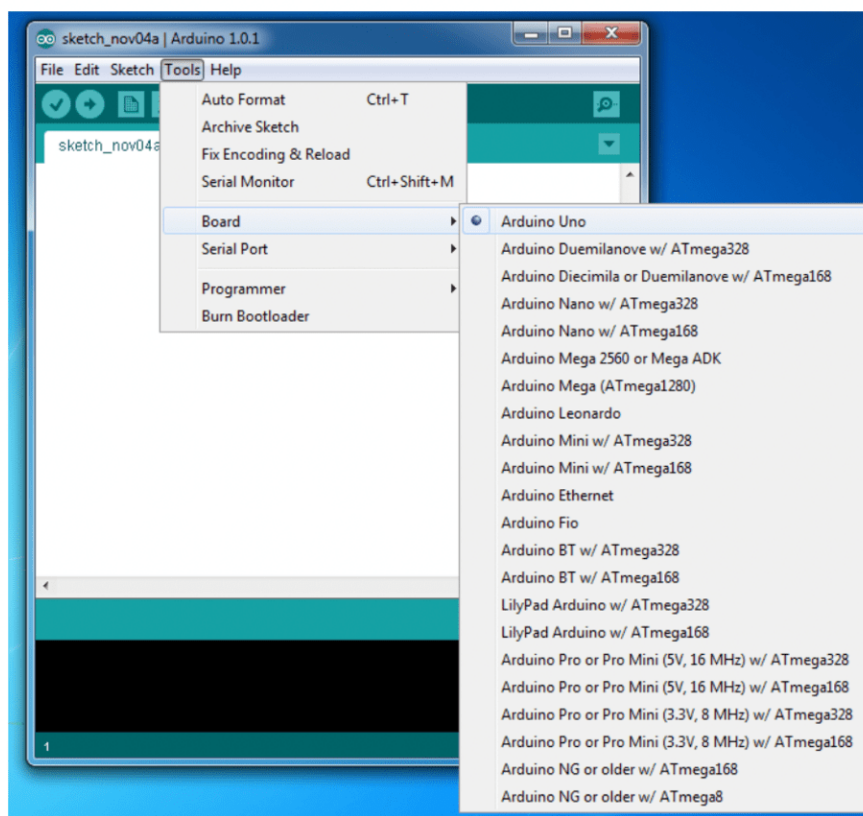
If you used the Installer, it'll install drivers automatically as soon as you connect your board.

4. Connect the board to your computer via the USB cable

To power up your board, connect your Arduino board with the pc via USB cable. The green color power LED should glow on the board.

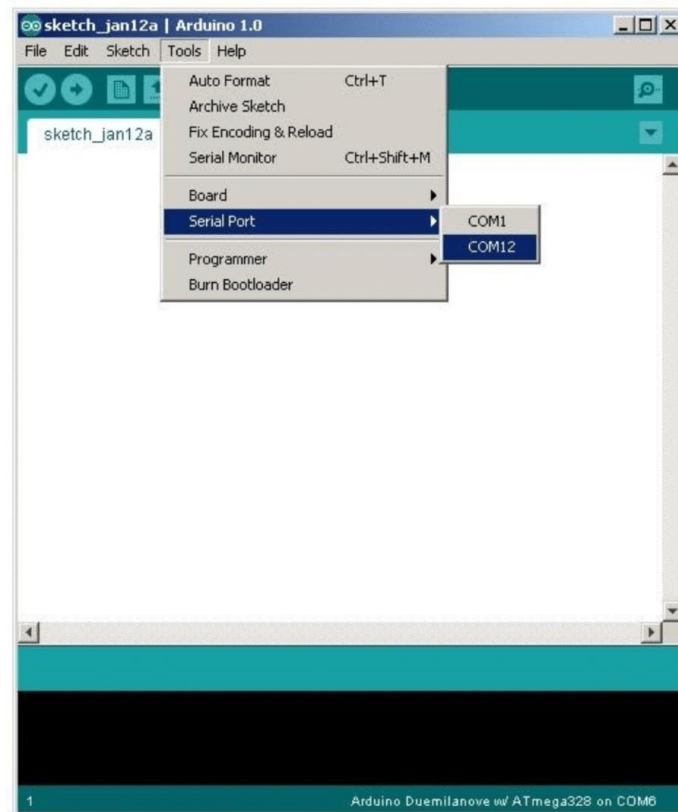
5. Select your board

Next, make sure the software is ready up for your particular Arduino board. Go to the “Tools” computer menu from the menu bar. Select the “Board” option and another menu will appear, where you'll select your Arduino model from the list.



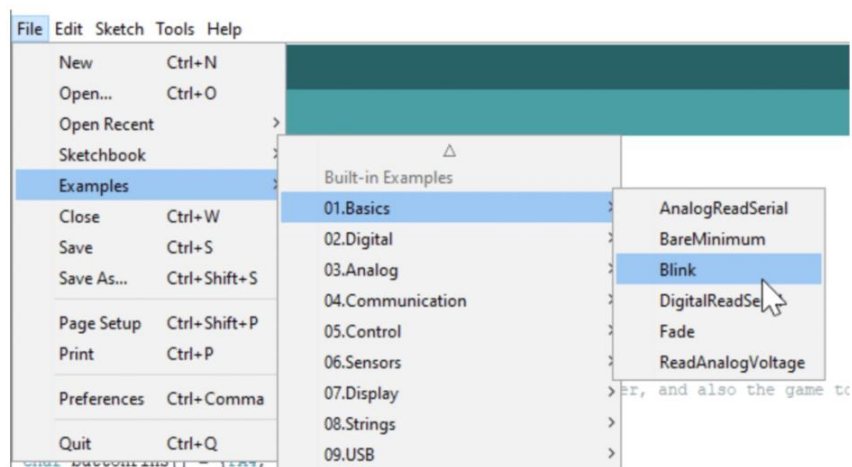
6. Select your serial port

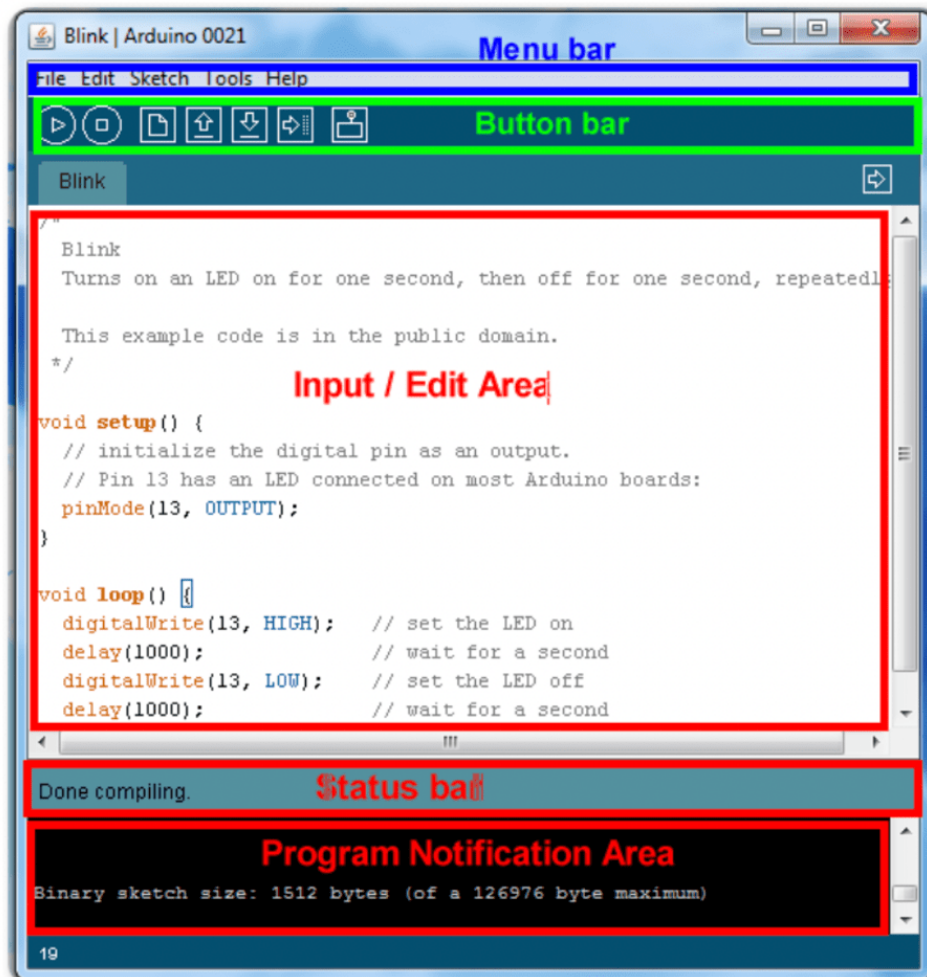
Select the serial device of the Arduino board. Go to Tools, and then the serial port menu. You might see COM3 or higher (COM1 and COM2 are usually reserved for hardware serial ports). To find out which port your Arduino board is connected to, disconnect your Arduino board and re-open the menu. The entry that disappears should be the Arduino board. Reconnect the board and choose that serial port.



7. Open the blink example

We'll start with the LED Blink example that comes with the Arduino IDE. Just go to File->Examples->Basics->Blink.



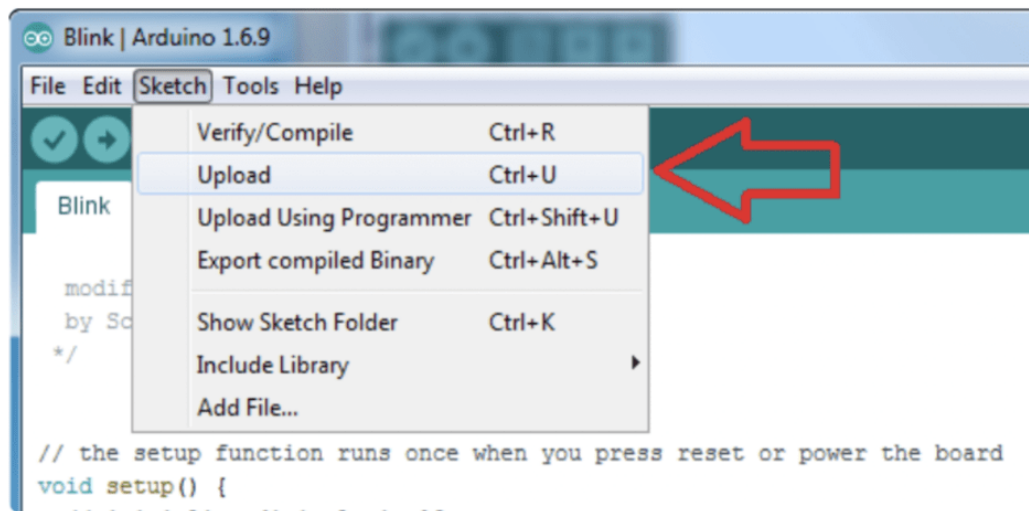


Here are the few things to keep in mind while writing the code:

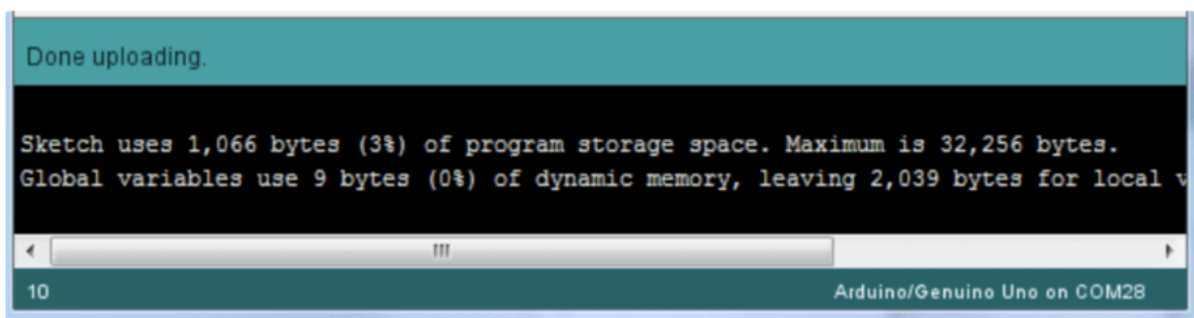
- Code is case sensitive
- All the statements must end with a semicolon
- Comments follow a `//` or begin with `/*` and end with `*/`
- `void loop()` and `void setup()` are two mandatory functions. The setup section of the code is simply run once when the Arduino board is first turned on or reset. Once the setup is complete, the loop runs over and over. It keeps on running until the board continues to stay powered.
- The status bar shows that the program is compiled or uploaded.
- Program notification area shows error(s) within the code if any.

8. Upload the program

Now it is time to upload your first sketch(code). Confirm the Arduino is plugged in, and the green light is on - therefore the correct board and port is chosen. Select Upload from the Sketch drop-down menu.



After a few seconds, you will get this screen, with the message "Done uploading."



5.1.3 Applications of Arduino

- 1) **Home Automation:** Arduino boards combined with sensors, actuators and wireless connectivity allows home owners to control various aspects of their living spaces with ease. It can be used in smart lighting, temperature control, automated security systems, energy management.
- 2) **Robotics:** With Arduino, you can integrate sensors, motors and controllers to create autonomous machines capable of performing specific tasks. From simple line-following robots to complex humanoid models, Arduino provides a robust foundation for learning and experimenting with robotics concepts.
- 3) **Wearable Technology:** Wearable devices such as smartwatches, fitness trackers and garments embedded with interactive elements can be created with the help of Arduino due to its low compact size, low power consumption and compatibility with various sensors and displays.
- 4) **Educational Teaching:** Arduino's accessibility and affordability makes it an excellent educational tool for teaching electronics, programming and problem-solving skills. Arduino-based projects provide hands-on learning experiences, encouraging students to explore technology and to develop a deep understanding of concepts.
- 5) **Automated Gardening:** Automated gardening is an increasingly popular way to grow plants with Arduino. This can be done by using digital sensors and actuators that interact with a user's

environment, such as temperature or water levels. This technology could be used for plant monitoring and controlling irrigation systems in greenhouses or large fields of crops.

- 6) **Smart City Projects:** Arduino is a powerful tool for enabling the development of smart city projects, such as automated parking systems, traffic control solutions or even energy-efficient lighting networks.
- 7) **Drones:** Arduino's ease of use allows users to customize each component of drone according to their exact needs and it also provides support for wireless communication protocols like Wi-Fi or Bluetooth. Custom drones for specific applications such as agriculture monitoring, search and rescue operations, etc. can be built.

5.2 ARCHITECTURE, PROGRAMMING & APPLICATIONS OF RASPBERRY PI

Introduction:

Raspberry Pi, developed by Raspberry Pi Foundation in association with Broadcom, is a series of small single-board computers and perhaps the most inspiring computer available today.

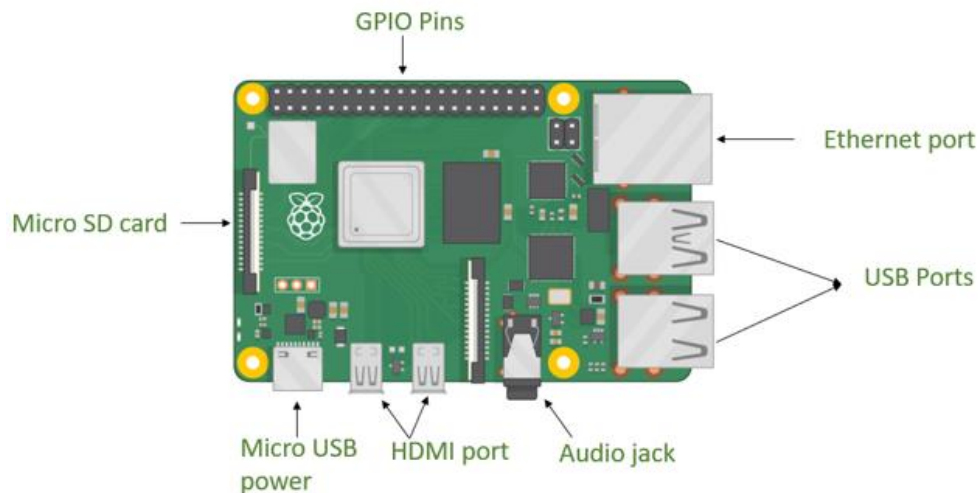
It is originally designed for education, inspired by the 1981 BBC Micro. Creator Eben Upton's goal was to create a low-cost device that would improve programming skills and hardware understanding at the pre-university level. The Raspberry Pi is slower than a modern laptop or desktop but is still a complete Linux computer and can provide all the expected abilities that implies, at a low-power consumption level.

Version Name	Year Released
Raspberry Pi 1 Model B	2012
Raspberry Pi 1 Model A	2013
Raspberry Pi 1 Model B+	2014
Raspberry Pi 2 Model B	2015
Raspberry Pi Zero	2015
Raspberry Pi 3 Model B	2016
Raspberry Pi Zero W	2017
Raspberry Pi 3 Model B+	2018
Raspberry Pi 3 Model A+	2018
Raspberry Pi 4 Model B	2019
Raspberry Pi 400	2020
Raspberry Pi Pico	2021
Raspberry Pi Zero 2 W	2021
Raspberry Pi 5	2023

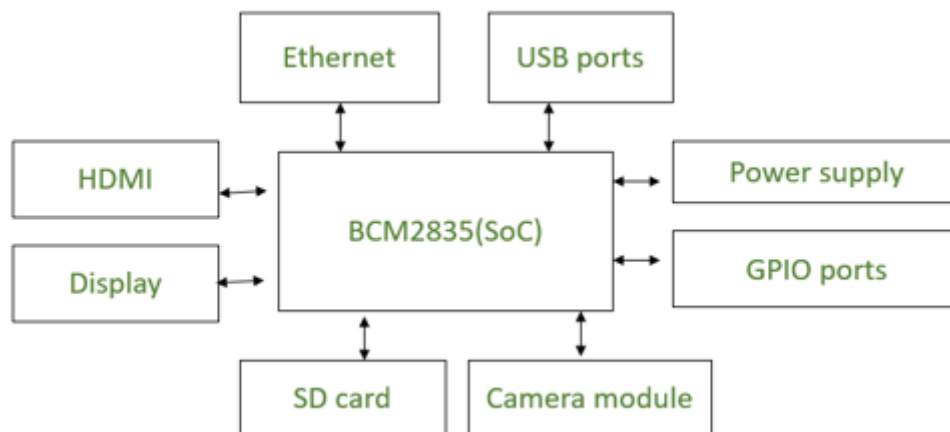
5.2.1 Architecture of Raspberry Pi

Raspberry Pi is a small single-board computer (SBC). It is a credit card-sized computer that can be plugged into a monitor. It acts as a minicomputer by connecting the keyboard, mouse and display. Raspberry Pi has an ARM processor and 512MB of RAM.

The following diagram shows the architecture of Raspberry Pi:



The following diagram shows some main blocks of Raspberry Pi:



Raspberry Pi mainly consists of the following blocks:

- 1) **Processor:** Raspberry Pi uses Broadcom BCM2835 system on chip which is an ARM processor and Video core Graphics Processing Unit (GPU). It is the heart of the Raspberry Pi which controls the operations of all the connected devices and handles all the required computations.
- 2) **HDMI:** High Definition Multimedia Interface is used for transmitting video or digital audio data to a computer monitor or to digital TV. This HDMI port helps Raspberry Pi to connect its signals to any digital device such as a monitor digital TV or display through an HDMI cable.
- 3) **GPIO ports:** General Purpose Input Output ports are available on Raspberry Pi which allows the user to interface various I/P devices.
- 4) **Audio output:** An audio connector is available for connecting audio output devices such as headphones and speakers.

- 5) **USB ports:** This is a common port available for various peripherals such as a mouse, keyboard, or any other I/P device. With the help of a USB port, the system can be expanded by connecting more peripherals.
- 6) **SD card:** The SD card slot is available on Raspberry Pi. An SD card with an operating system installed is required for booting the device.
- 7) **Ethernet:** The ethernet connector allows access to the wired network, it is available only on the model B of Raspberry Pi.
- 8) **Power supply:** A micro USB power connector is available onto which a 5V power supply can be connected.
- 9) **Camera module:** Camera Serial Interface (CSI) connects the Broadcom processor to the Pi camera.
- 10) **Display:** Display Serial Interface (DSI) is used for connecting LCD to Raspberry Pi using 15 15-pin ribbon cables. DSI provides a high-resolution display interface that is specifically used for sending video data.

5.2.2 Applications of Raspberry Pi

- 1) Desktop PC
- 2) Wireless print server
- 3) Media Usage
- 4) Game Servers
- 5) Retro Gaming Machine
- 6) Robot Controller
- 7) Stop Motion Camera
- 8) Time-lapse Camera
- 9) FM Radio Station
- 10) Web Servers
- 11) Playing games
- 12) Browsing the internet
- 13) Word processing
- 14) Spreadsheets
- 15) Editing photos
- 16) Paying bills online
- 17) Managing your accounts

5.3 RASPBERRY PI V/S ARDUINO

Parameters	Arduino	Raspberry Pi
Control Unit	The Control Unit of the Arduino is from the ATmega family.	The Control Unit of the Raspberry Pi is from the ARM family.
Basis	Arduino works on the basis of a microcontroller.	Raspberry Pi, on the other hand, works on the basis of a microprocessor.
Use	The Arduino basically helps in controlling all the electrical components that connect to a system's circuit board.	The Raspberry Pi primarily computes data and info for producing valuable outputs. It also controls the various components in any given system on the basis of the outcome (of the computation).
Structure of Hardware and Software	The Arduino boards have a very simple structure of software and hardware.	The Raspberry Pi boards consist of comparatively complex software and hardware architecture.
Type of CPU Architecture	Arduino has an 8-bit architecture.	Raspberry Pi has a 64-bit architecture.
RAM Usage	Arduino makes use of very little RAM of about 2 kB (Kilobytes).	Raspberry Pi always requires more RAM than Arduino of about 1 GB (Gigabytes).
Processing Speed	Arduino clocks 16 MHz (Megahertz) of processing speed in a system.	The Raspberry Pi clocks 1.4 GHz (Gigahertz) of processing speed in a system.
Cost Efficiency	It has a higher cost-efficiency because it is comparatively cheaper.	It has a lower cost-efficiency because it is comparatively more expensive.
I/O Drive Strength	The I/O current drive strength in the case of Arduino is higher.	The I/O current drive strength in the case of Raspberry Pi is lower.
Power Consumption	Arduino consumes power of about 200 MW (Megawatts).	Raspberry Pi consumes about 700 MW.
Software	Arduino boards are programmable using C/C++ languages.	Raspberry Pi supports its own Linux-based operating system Raspberry Pi OS. You can also install the OS you like.
Internet	Arduino does not have internet support. You need additional modules or shields to connect it to the internet.	Raspberry Pi has a built-in Ethernet port and WiFi support.

INTERNET OF THINGS

Current drive strength	Higher current drive strength	Lower current drive strength
Capability	Arduino is generally used to perform single (and simple) tasks repeatedly.	Raspberry Pi can perform multiple tasks simultaneously.
Wireless connectivity	Arduino does not support Bluetooth or WiFi.	Raspberry Pi supports Bluetooth and WiFi.
Applications	Traffic light countdown timer, Parking lot counter, Weighing machines, etc.	Robot controller, Game servers, Stop motion cameras, etc.